

LIGHTMATTER CONFIDENTIAL

Silicon Photonic Waveguide Step-Function Response and SSPRQ Thermal Wander

2D Finite-Difference Simulation · September 30, 2026

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KEY RESULT — Caribou-MA-DR MRM (PAM4, 106.25 GBaud)

Convolution of the full IEEE 802.3 SSPRQ PAM4 pattern (65,535 symbols at 106.25 GBaud) with the thermal impulse response, after 100 precondition cycles to reach periodic thermal steady state, yields a **peak-to-peak thermal wander of 352.1 mK (23.59 pm, 91.90% of one PAM level separation)**, with sigma = 65.74 mK (4.404 pm, 17.16% of one PAM level separation) and a mean offset of **+0.01 mK (+0.001 pm, +0.00% of one PAM level separation)** which is the residue of SSPRQ's odd symbol count, not a DC imbalance (see Appendix D). This is a **signal-integrity concern that must be considered for TDECQ and BER simulations**.

KEY RESULT — Caribou-MA-OCI MRM (NRZ, 106.25 GBaud)

Convolution of the full SSPR NRZ pattern (32,762 bits at 106.25 GBaud, $T_{\text{cycle}} = 308.3$ ns) with the same thermal impulse response (identical Si cross-section), after 100 precondition cycles, yields a **peak-to-peak thermal wander of 394.9 mK (26.46 pm, 31.88% of the 83 pm NRZ OMA swing)**, with sigma = 96.69 mK (6.478 pm, 7.81% of the OMA swing). Unlike SSPRQ, SSPR is **exactly DC balanced**, so the mean offset is exactly zero and there is no static thermal bias to budget. The cost of that construction is a large low-frequency excursion instead: SSPR's second half is the bitwise inverse of its first, which places a strong tone at 3.24 MHz, inside the thermal passband. This is a **signal-integrity concern that must be considered for TDEC and BER simulations**.

1. Overview

This report characterises the **step-function thermal response** and **pattern-dependent thermal wander** for two silicon microring modulator use cases, each driven by its own standard stress pattern — SSPRQ for the PAM4 DR case and SSPR for the NRZ OCI case — and sharing an identical 370x270 nm Si waveguide cross-section in the W via-fill contact geometry (see Section 2):

- **Caribou-MA-DR MRM** — PAM4, 106.25 GBaud, radius 9.815 μm (circumference 61.67 μm), $\text{OMA}_{\text{outer}} = 5.767 \text{ dBm} = 3.773 \text{ mW}$ (Caribou_OMA_rollup_2026-08-31, use case 80, cell H36, median case).
- **Caribou-MA-OCI MRM** — NRZ, 106.25 GBaud, radius 6.28 μm (circumference 39.46 μm), $\text{OMA} = 3.793 \text{ dBm} = 2.395 \text{ mW}$ (83 pm swing, Caribou_OMA_rollup_2026-08-31, use case 70, cell H40).

Both MRMs share the identical W via-fill cross-section: a W + SiO_2 composite (16 % W via-fill, 84 % SiO_2 by volume, $k = 25.93 \text{ W/(m}\cdot\text{K)}$) occupies the region from 2.5 μm outboard of the core edge to the lateral domain boundary, above the core top (270 nm) up to an upper heatsink 0.5 μm above the core. Because the cross-section is the same, the normalised thermal impulse response $h(t) [\text{K}\cdot\text{m}/(\text{W}\cdot\text{s})]$ is shared; only the ring circumference and OMA differ between the two use cases.

All temperatures in this document are the **spatial mean over the 370x270 nm Si waveguide core**. The guided optical mode is confined within the core, and the high thermal conductivity of Si (85 $\text{W/(m}\cdot\text{K)}$) keeps intra-core gradients negligible on all timescales of interest.

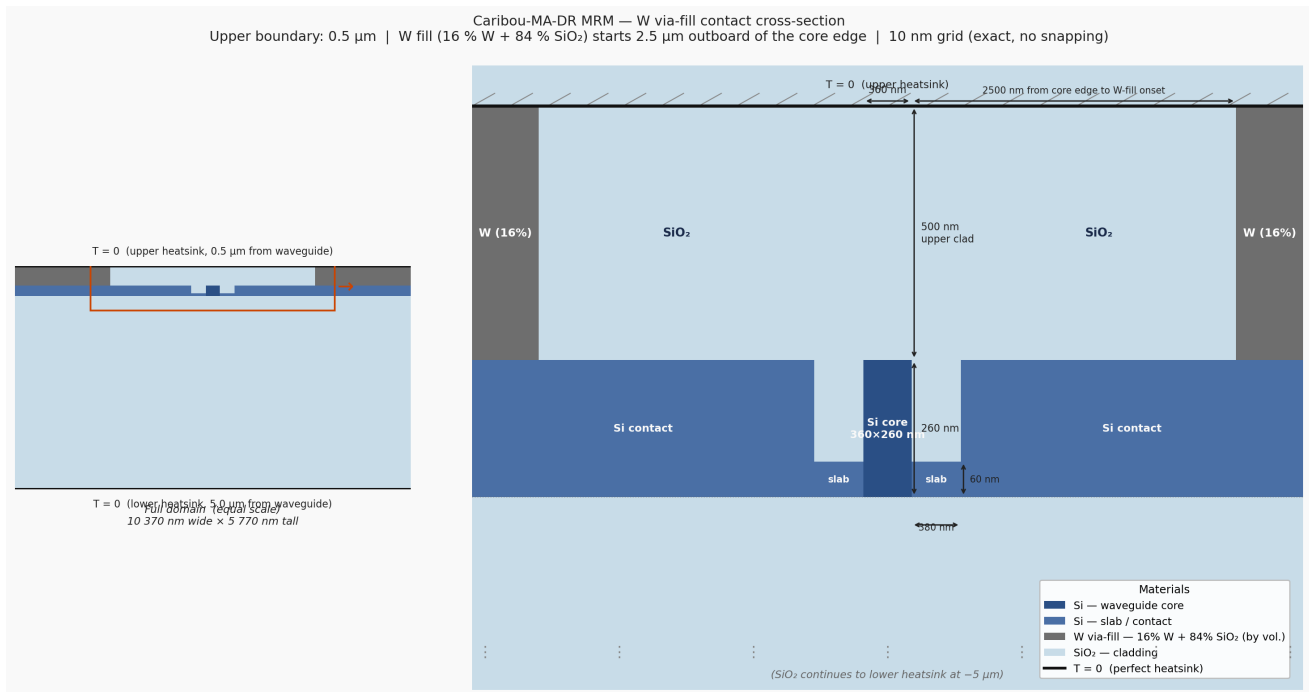
Caribou-MA-DR: The peak uncompensated temperature rise is 1.9996 K (a model-internal normalisation, not a device prediction - C.2) (133.97 pm resonance shift) for an $\text{OMA}_{\text{outer}}/2$ step. Convolution of the full IEEE 802.3 SSPPQ PAM4 pattern (65,535 symbols) with the thermal impulse response, after 100 precondition cycles, yields a **peak-to-peak thermal wander of 352.07 mK (23.589 pm, 91.90% of one PAM level separation)**. 90 % of the step-function steady state is reached in 578.7 ns. The W-fill path dominates the heat removal: 95 % of the steady-state heat leaves through the 0.5 μm upper boundary rather than the 5 μm substrate path.

Caribou-MA-OCI: The stress pattern is SSPPR, a single 32,762-bit NRZ stream ($T_{\text{cycle}} = 308.3 \text{ ns}$). Convolution with the same impulse response (identical cross-section, OCI power per unit length scaled by ring circumference) after 100 precondition cycles yields a **peak-to-peak thermal wander of 394.91 mK (26.459 pm, 31.88% of the 83 pm NRZ OMA swing)**, with a mean offset of exactly zero because SSPPR is exactly DC balanced. **Measured against the eye each format must fit inside, DR is the more constrained of the two:** its excursion is 91.9% of one PAM level against OCI's 31.9% of the NRZ swing, so although OCI is the larger in millikelvin it has roughly three times the room to absorb it (5.4.3). The thermal time constants are identical to the DR case (shared cross-section); the SSPPR cycle period (308 ns) is comparable to the 578.7 ns thermal 90% rise time, placing the wander squarely in the thermally-relevant regime.

Mathematical details are in Appendix A. Modelling assumptions are in Appendix C. Pattern DC balance and preconditioning are in Appendix D.

2. Geometry Cross-Section — Caribou-MA-DR and Caribou-MA-OCI MRMs

Both MRM use cases share the identical waveguide cross-section shown below. Only the ring radius (and therefore circumference and absorbed power per unit length) differs between Caribou-MA-DR ($R = 9.815 \mu\text{m}$) and Caribou-MA-OCI ($R = 6.28 \mu\text{m}$).



3. Key Results

Read the steady-state rows below as model normalisation, not as device predictions. The boundary conditions were placed to resolve the near field of the waveguide, which is what governs the pattern-dependent wander this report is about. They do not represent the real structure further out, and the part is undercut where this model is not. The steady-state thermal resistance that follows, 1.06 K/mW for the DR ring, is therefore low by roughly a factor of six against the foundry's measured 6.2 K/mW — and the foundry's figure is the one to use. Everything marked below inherits that error: ΔT_{ss} , the static resonance shift, and the rise times beyond about 25 ns. The **pattern-dependent wander is much less exposed**, because a more isolated ring raises the thermal resistance and lowers the roll-off corner by nearly the same factor, so the response where the patterns actually put their energy barely moves. C.2 gives the timescale analysis and the evidence.

3.1 Caribou-MA-DR (PAM4, 106.25 GBaud)

Quantity	Value
OMA _{outer} (PAM 0 to PAM 3)	5.767 dBm = 3.773 mW [use case 80, cell H36, median case]
Ring radius / circumference	9.815 μm / 61.67 μm

Initial T_wg (controller set point)	1.9996 K (OMA_outer/2 steady state) [not predictive - see C.2]
Peak delta-T (midpoint -> PAM3)	1.9996 K (OMA_outer/2 step) [not predictive - see C.2]
Peak resonance shift	133.97 pm (67 pm/K x 1.9996 K)
PAM level spacing	25.667 pm (77 pm over 3 levels)
10 / 50 / 90% rise times	2.3 ns / 50.1 ns / 578.7 ns
SSPRQ wander (pk-pk)	352.07 mK = 23.589 pm = 91.90% of 1 PAM level
SSPRQ wander (sigma)	65.74 mK = 4.404 pm = 17.16% of 1 PAM level
SSPRQ wander (mean offset)	+0.010 mK = +0.001 pm = +0.00% of 1 PAM level (SSPRQ is DC balanced; D.1)

3.2 Caribou-MA-OCI (NRZ, 106.25 GBaud)

Quantity	Value
OMA (bit 0 to bit 1)	3.793 dBm = 2.395 mW [use case 70, cell H40]
Ring radius / circumference	6.28 μ m / 39.46 μ m
NRZ OMA swing (bit 0 -> 1)	83 pm
dT_ss (analytical from DR)	1.9837 K (same cross-section, scaled power) [not predictive - see C.2]
Peak resonance shift	132.9 pm = 160.1% of OMA swing
10 / 50 / 90% rise times	2.3 ns / 50.1 ns / 578.7 ns (identical cross-section)
Test pattern	SSPR, 32,762 bits, T_cycle = 308.35 ns
SSPR wander (pk-pk)	394.91 mK = 26.459 pm = 31.88% of OMA swing
SSPR wander (sigma)	96.69 mK = 6.478 pm = 7.81% of OMA swing
SSPR wander (mean offset)	+0.000 mK = +0.000 pm = +0.00% of OMA swing (exactly zero: SSPR is DC balanced)
SSPR wander (min / max)	-197.45 / +197.45 mK (symmetric about zero)

3.3 Neither Distribution Is Gaussian — Do Not Q-Factor sigma

Both wander distributions are reported above with a sigma, and the figures overlay a normal fit for reference. That fit is a scale indicator, not a model: **neither distribution is Gaussian, and in both cases sigma overstates the tail risk.** Taking either sigma and extrapolating a bit-error tail through a Q-factor predicts excursions that never occur, because both distributions are hard-bounded well inside where a Gaussian of the same sigma would reach.

Shape metric	DR (SSPRQ)	OCI (SSPR)	Gaussian
Skew	-0.33	+0.00	0
Excess kurtosis	+0.88	-0.70	0
Fraction beyond 2 sigma	2.24x Gaussian	0.03x Gaussian	1.00x
Fraction beyond 3 sigma	0.00x Gaussian	0.00x Gaussian	1.00x
Peak-to-peak / sigma	5.36	4.08	~8.6 / ~8.3

DR (SSPRQ) is peaked with heavy shoulders, but no tails. It is negatively skewed (-0.33) and leptokurtic (+0.88), and at 2 sigma it is 2.24 times as populated as a Gaussian — yet *no* sample in the cycle reaches 3 sigma at all. Its whole peak-to-peak span is 5.36 sigma, against the ~8.6 sigma a Gaussian sample of this length would cover. Excess population in the shoulders with nothing beyond them is what a bounded deterministic waveform looks like, and it is not what a positive kurtosis alone would lead you to expect.

OCI (SSPR) is flat-topped, and bounded more tightly still. It is platykurtic (-0.70) and symmetric by construction (Section 5.4.2), and *no* sample in the cycle falls beyond 3 sigma. Its whole peak-to-peak span is only 4.08 sigma, against the ~8.3 sigma a Gaussian sample of this length would span. The distribution is not a sum of many small independent contributions — it is dominated by one large, deterministic, periodic swing, which is why it has shoulders rather than tails. The same conclusion holds for DR: in both cases the peak-to-peak is a hard bound recurring every cycle, not a rare event, and it is the figure to carry into a margin calculation. The two differ in *shape* — DR peaked, OCI flat-topped — not in which side of Gaussian their tails fall on.

The practical rule for both cases is the same: this wander is deterministic and pattern-locked, not random noise. It repeats every pattern cycle. It should enter TDECQ, TDEC and BER simulation as a time-correlated distortion driven by the actual pattern, not as an additional Gaussian noise term with the sigma quoted above.

3.4 The Same Result Seen at the Optical Output

Sections 3.1 and 3.2 give the wander as a temperature and a resonance shift. The figure below restates it as the quantity a link engineer reads directly: the optical power leaving the modulator, and how far each level drifts from where it is supposed to sit. The conversion is the one used throughout — a thermal resonance shift of $\Delta\lambda = \Delta T \times 67 \text{ pm/K}$ is indistinguishable from a drive error of the same size, so dividing by the 25.67 pm spacing of a DR PAM level, or the 83 pm of an OCI NRZ level, gives the level error directly.

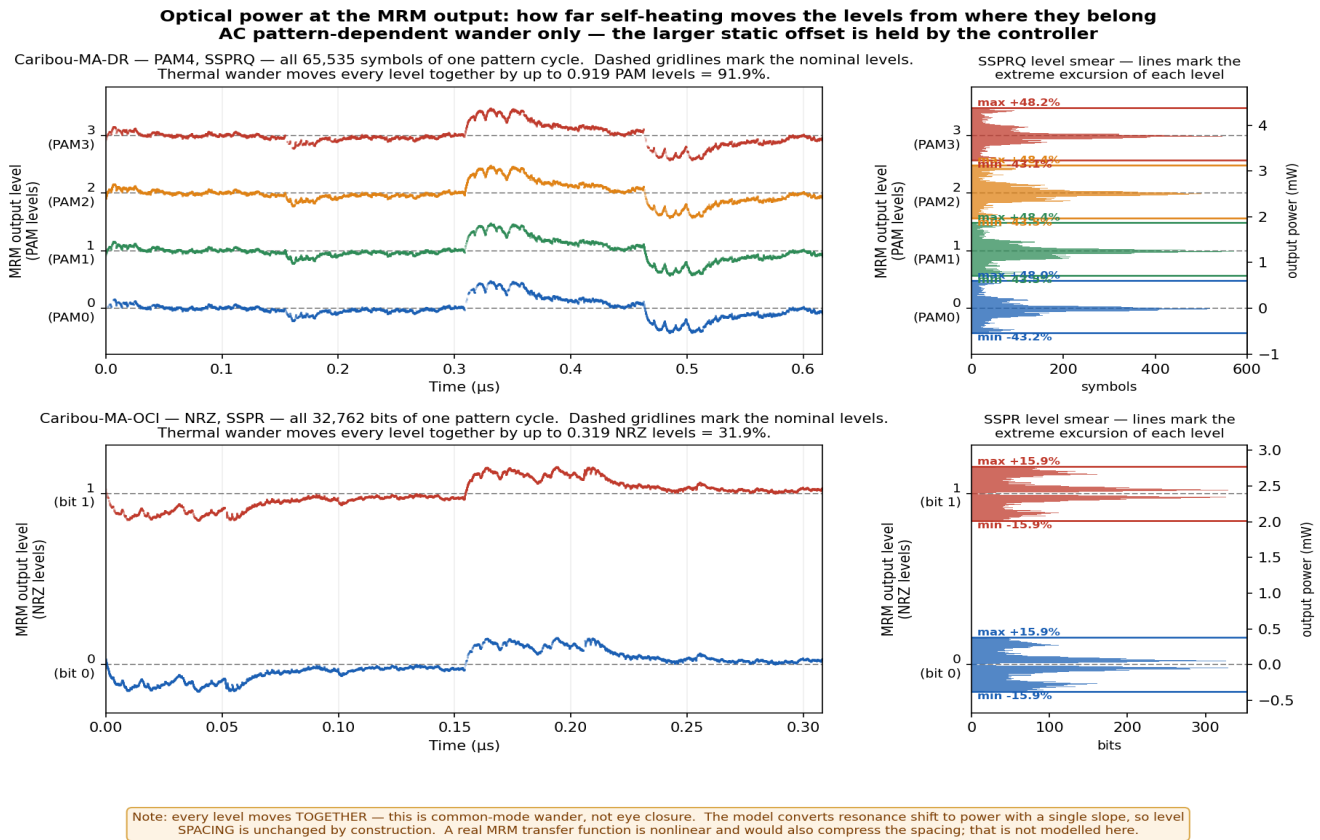


Figure 2 — MRM output level against time, for every symbol of one pattern cycle. Left panels: the level of each transmitted symbol, with dashed gridlines on the nominal levels. Right panels: the resulting smear of each level, with solid lines marking its extreme excursion. Top row Caribou-MA-DR under SSPRQ, bottom row Caribou-MA-OCI under SSPR. This is the AC pattern-dependent wander about the controller set point; the much larger static offset is held by the controller and is not shown (C.1, C.8).

Two features stand out that the temperature plots do not make obvious. **The DR excursion is mildly asymmetric** — about +48.4% and -43.5% of a PAM level, a ratio of 1.11 between the two sides. That is the skew of the SSPRQ distribution (3.3) showing up at the output, with the sign reversed: heating moves the output *down*, so a delta-T distribution skewed -0.33 appears as a level distribution leaning the other way. The asymmetry is small enough that a margin taken as plus-or-minus half the peak-to-peak would be only slightly wrong, but the peak-to-peak itself is 92% of a PAM level and that is the number that matters. **The OCI excursion is exactly symmetric**, at +15.9% and -15.9% of the NRZ swing, which is the half-wave antisymmetry of SSPR (5.4.2) appearing once more — the same property that gives that pattern exactly zero mean and a spectrum with every even harmonic nulled.

One caveat on how to read the figure. The model converts a resonance shift into an output power with a single slope, so the thermal shift displaces every level by the same amount and leaves the spacing between them untouched. The wander shown is therefore entirely common-mode, and the figure is not an eye-closure plot. A real MRM transfer function is nonlinear, so a resonance shift would also compress and expand the level spacing; that differential component is not modelled here and would have to be added before these amplitudes are turned into an eye penalty.

4. Caribou-MA-DR — Full Analysis

Sections 4.1 – 4.6 present the complete FEM simulation, step-function response, thermal frequency response, physical interpretation, and SSPRQ PAM4 thermal wander analysis for the Caribou-MA-DR MRM. The OCI analysis follows in Section 5.

4.1 Simulation Parameters

Parameter	Value
Device	Caribou-MA-DR-MRM, radius 9.815 μm , circumference 61.67 μm ($2\pi \times 9.815 \mu\text{m}$)
Waveguide cross-section	370 nm x 270 nm (Si core); 70 nm shoulder slabs, 380 nm wide
Cladding — bottom / sides	SiO_2 , 5 μm
Cladding — top	SiO_2 , 0.5 μm (upper heatsink boundary)
W-fill composite	16 % W via-fill + 84 % SiO_2 (by volume); $k = 25.93 \text{ W/(m}\cdot\text{K)}$, $\rho = 4928 \text{ kg/m}^3$, $C_p = 429.2 \text{ J/(kg}\cdot\text{K)}$. Region: 2.5 μm outboard of the core edge, y from core top (270 nm) to upper boundary (770 nm).
Outer boundary condition	$T = 0 \text{ K}$ (perfect heatsink at upper and lower cladding edges)
OMA_outer (PAM3 - PAM0)	5.767 dBm = 3.773 mW total (61.2 mW/mm) [Caribou_OMA_rollup_2026-08-31, use case 80, cell H36]
PAM level spacing	25.667 pm (77 pm total, 3 equal intervals)
Initial condition	Steady state at $\text{OMA_outer}/2 = 1.887 \text{ mW}$ (30.6 mW/mm); $T_{\text{wg_init}} = 1.9996 \text{ K}$. Controller stabilisation time assumed: 10 ms.
Power step simulated	$\text{OMA_outer}/2 = 1.887 \text{ mW}$ (30.6 mW/mm) (midpoint initial condition to PAM4 level 3)
Operating wavelength	1310 nm
FSR (measured)	6.65 nm
Group index n_g	4.18 (from FSR and ring radius)
Wavelength shift / ΔT	67 pm/K (67 pm/K — same for DR and OCI; material/wavelength property)
Operating temperature T_{op}	150 $^{\circ}\text{C}$ (423.15 K) — all material properties are evaluated here, not at 300 K (Appendix C.4)

Grid spacing	10 nm x 10 nm (exact fit to every feature; no snapping)
Time-domain range	1 ps to 20 ms (decade-blocked backward Euler, 200 steps/decade)
Model geometry	2D cross-section only — uniform heating along ring assumed; axial conduction, actual etch profile, and nearby metal interconnect layers not included (see Appendix C)

All material properties are evaluated at the operating temperature, $T_{op} = 150\text{ }^{\circ}\text{C}$, not at room temperature. The two silicon columns differ only in doping: core and slab are below $1\text{e}+18\text{ cm}^{-3}$, the contacts are taken at $5\text{e}+19\text{ cm}^{-3}$. Sources, temperature dependences and sensitivities are in Appendix C.4.

Material Property	Si core/slab	Si contact	SiO ₂	W-fill (16% W)
Thermal conductivity k (W/m·K)	85.3	65.5	1.53	25.93
Density (kg/m ³)	2,330	2,330	2,200	4,928
Specific heat C_p (J/kg·K)	800	800	916	429.2
Thermal diffusivity (m ² /s)	4.57e-05	3.51e-05	7.58e-07	1.23e-05

4.2 Step-Function Response

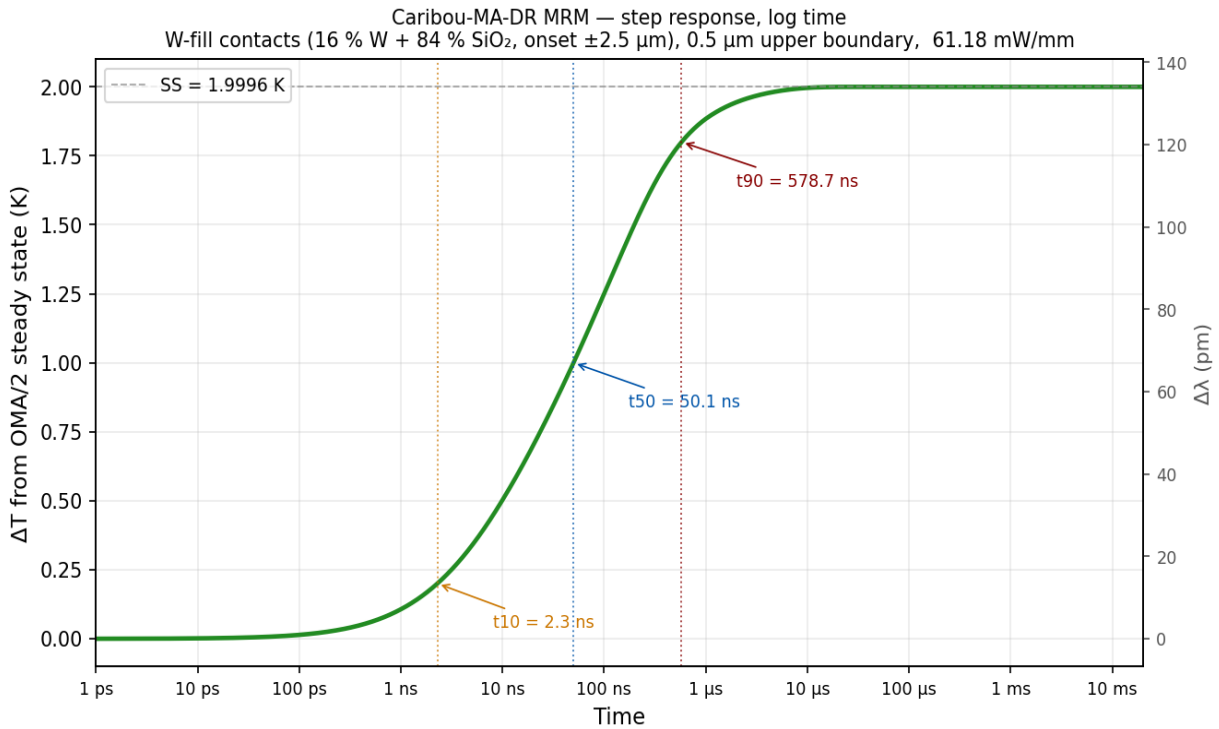


Figure 3. Step-function response, log time scale: spatial mean ΔT over the 370x270 nm Si waveguide core after a power step of OMA_{outer}/2 = 1.887 mW (30.6 mW/mm) from the midpoint initial condition ($T_{\text{wg_init}} = 1.9996$ K). Dashed line: peak $\Delta T = 1.9996$ K (133.97 pm resonance shift). Right axis: resonance shift in pm (67 pm/K). Dotted vertical lines mark $t_{10} = 2.3$ ns, $t_{50} = 50.1$ ns, $t_{90} = 578.7$ ns.

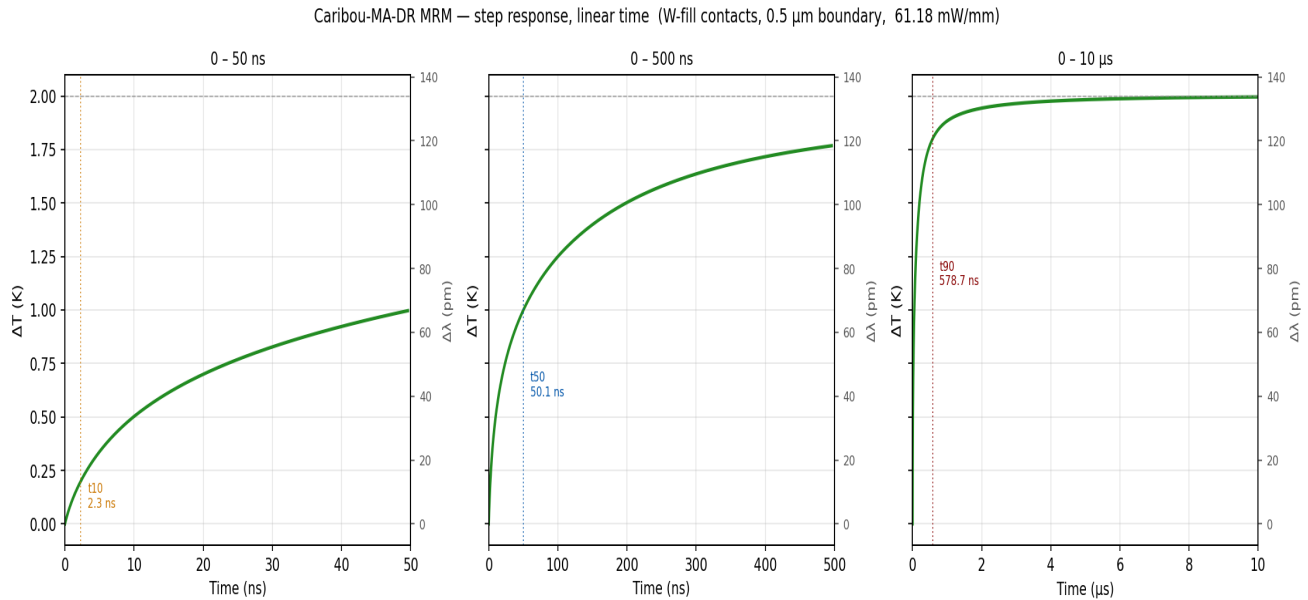
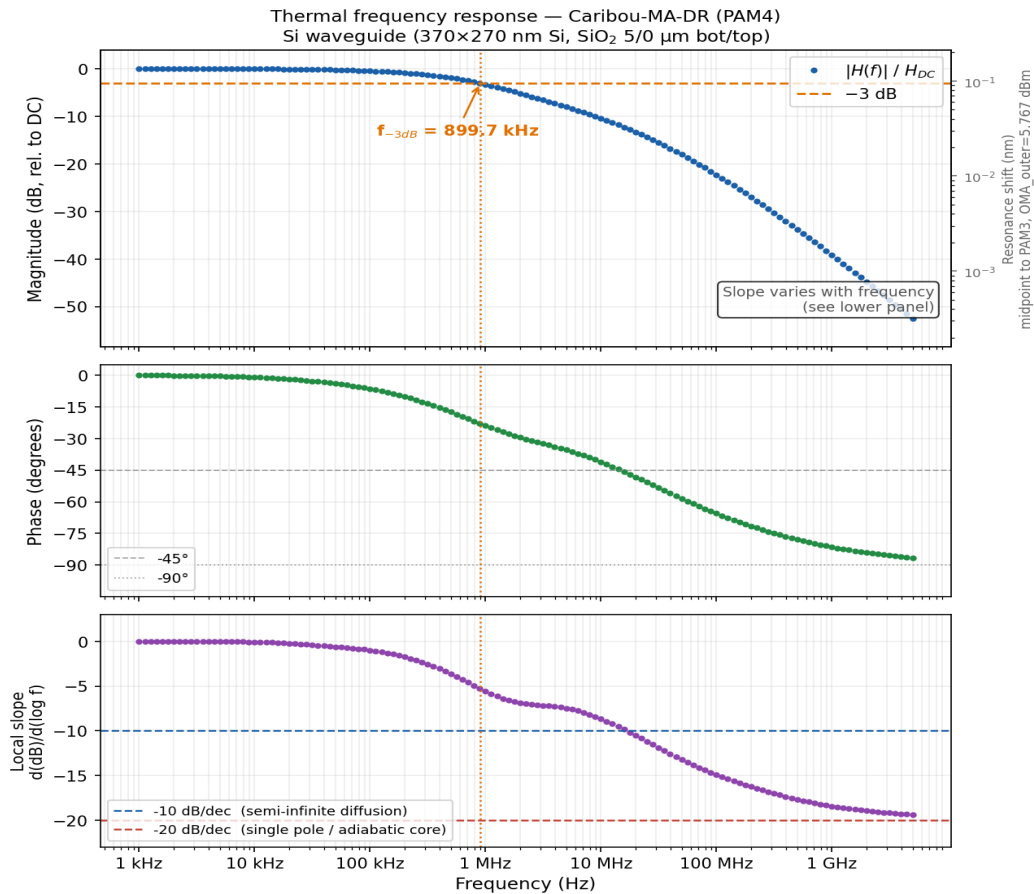


Figure 4. Step-function response, linear time: same data as Figure 3 shown over three successive time windows to reveal the thermal dynamics at each regime. Left (0–50 ns): rapid initial rise as the Si core equilibrates and heat enters the surrounding SiO₂; $t_{10} = 2.3$ ns and $t_{50} = 50.1$ ns are visible. Centre (0–500 ns): approach to the 90% rise time ($t_{90} = 578.7$ ns). Right (0–10 μs): full settling to the steady-state plateau (1.9996 K dashed). All panels show the spatial mean over the 370x270 nm Si core; right axis in pm resonance shift.

4.3 Thermal Frequency Response

The step response of 4.2 and the frequency response below are two views of the same system, and the pattern-dependent wander is easier to reason about in the second: a drive pattern is a spectrum, and what reaches the core is that spectrum multiplied by $H(f)$. The curve is obtained independently of 4.2, by solving the phasor equation $(j \cdot \omega \cdot D - L) \cdot T = Q$ directly at each frequency — one complex sparse factorisation per point, no fit and nothing interpolated between points (A.1).

This response belongs to the cross-section, not to either device. $H(f)$ is normalised per unit absorbed power per unit length, and Caribou-MA-DR and Caribou-MA-OCI have the same cross-section, so their normalised responses are *identical* — one phasor solve stands behind both. The corner frequency, the roll-off shape and H_{dc} quoted below therefore apply to the OCI device unchanged; the only device-specific quantity is the absolute rise $\Delta T_{ss} = H_{dc} \times P_{step}$. The same curve is reproduced as Figure 7 in 5.3, rescaled for OCI, so that section stands on its own. That two independent routes to this curve agree is verification rather than result, and is shown in B.5.



This is the SAME curve in both use-case sections — one phasor solve, not two. $H(f)$ is normalised per unit absorbed power and the DR and OCI devices share the cross-section exactly, so the normalised response is identical. Only the right-hand axis above, which converts it to an absolute resonance shift, is specific to this device.

Figure 5 — thermal Bode plot from the direct phasor solve. This curve is a property of the shared cross-section and applies to both use cases; Figure 7 is the same curve with the OCI scaling. Each marker is one independent complex solve; no line joins them, because nothing is computed between the points. The lower panel gives the LOCAL slope, which is the point of the figure: it is not constant, so the response is not a single pole. Note that the thermal -3 dB frequency lies BELOW the SSPRQ cycle fundamental, so essentially all of the pattern's spectral content is attenuated — yet the residual response still produces the wander reported in Section 4.6. Low-pass attenuation suppresses the wander amplitude; it does not eliminate it.

4.3.1 Frequency-Domain Results

Quantity	Value
DC thermal impedance H_{dc}	0.065364 K·m/W = 65.36 K·mm/W
-3 dB frequency	0.900 MHz (899.7 kHz)
Local rolloff slope	varies from -0.0 to -19.4 dB/decade across the sweep — no single value applies
Sweep	1 kHz to 5 GHz, 135 points

The rolloff is not a single slope, and this system is not a single pole. Thermal responses are often modelled as one pole, which would give a constant -20 dB/decade above one corner frequency. That is not what the solve shows:

Frequency	H (dB re DC)	Local slope (dB/decade)	Phase (deg)
99.9292 kHz	-0.44	-0.98	-6.5
998.939 kHz	-3.25	-5.58	-23.9
9.98585 MHz	-10.41	-8.67	-41.1
99.8232 MHz	-22.27	-14.93	-65.5
997.878 MHz	-39.17	-18.44	-81.4
5 GHz	-52.46	-19.38	-86.5

The slope climbs continuously over about four decades and has still not settled at the top of the sweep. Fitting the best single pole to the computed magnitude leaves a residual of **9.51 dB RMS and 17.95 dB at worst** (best-fit corner 1.51 MHz), so the single-pole approximation is not a small error — it is the wrong functional form. Any budget or control-loop argument that assumes one pole, or a fixed dB/decade, will misestimate the thermal response by many dB depending on the frequency of interest.

Physically there are three regimes. Below roughly 100 kHz the core follows the drive and the response is flat. Through the middle decades heat diffuses into the cladding and the behaviour is distributed rather than lumped, the slope passing through the -10 dB/decade characteristic of semi-infinite diffusion (H proportional to $1/\sqrt{j \cdot \omega}$). At the top of the sweep the thermal penetration depth becomes small compared with the core itself, heat has no time to leave during a cycle, and the response approaches that of the core's own heat capacity alone — ΔT proportional to $Q/(j \cdot \omega \cdot \rho \cdot C_p)$, giving -20 dB/decade and -90 degrees. The -20 dB/decade reached at the top of this sweep is that adiabatic limit, not evidence of a pole.

4.4 Physical Interpretation

The step-function response shows three regimes driven by the material contrast in thermal diffusivity: Si ($4.57e-05 \text{ m}^2/\text{s}$), SiO_2 ($7.58e-07 \text{ m}^2/\text{s}$), and the W-fill composite ($1.23e-05 \text{ m}^2/\text{s}$):

- **Sub-nanosecond (~399 ps):** The Si core equilibrates. The early rapid rise (0.2000 K, 13.4 pm by 2.3 ns) reflects heat entering the immediately surrounding SiO_2 .

- **Nanosecond to 578.7 ns:** Heat diffuses upward through the $0.5\ \mu\text{m}$ SiO_2 cladding above the core (direct-path time constant $\sim 82\ \text{ns}$) and spreads laterally. As heat reaches the W-fill composite $2.5\ \mu\text{m}$ outboard of the core edge, the higher diffusivity ($1.23\text{e-}05\ \text{m}^2/\text{s}$) routes it rapidly to the upper heatsink, shortening the thermal tail. 90% of steady state is reached by 578.7 ns.
- **Steady state:** Peak $\Delta T = 1.9996\ \text{K}$ (133.97 pm resonance shift) for the OMA_outer/2 step. Ring thermal resistance 1059.9 K/W (65.36 K/(W/mm)). Response scales linearly with absorbed power.

4.5 SSPRQ Pattern-Dependent Thermal Wander — Caribou-MA-DR

The thermal impulse response $h(t)$ [K·m/(W·s)] was convolved with the absorbed-power sequence $P_{ac}(t)$ derived from the full IEEE 802.3 SSPRQ PAM4 pattern (65,535 symbols at 106.25 GBaud, $T_{sym} = 9.41$ ps). SSPRQ is the standard stress pattern for PAM4 transceiver qualification: it is built from three PRBS31 sections (Table 120-2) Gray-coded into PAM4, with two of the four resulting symbol blocks inverted, to maximise disparity run lengths and worst-case inter-symbol interference. For the Caribou-MA-DR MRM through port, absorbed power at each PAM level is: level 0 = 3.773 mW (maximum absorption), level 1 = 2/3 OMA, level 2 = 1/3 OMA, level 3 = 0 mW. The AC component $P_{ac} = P - OMA/2$ (deviation from the balanced mean) drives the thermal wander $\Delta T = h * P_{ac}$, computed via FFT convolution. SSPRQ is DC balanced — the running disparity returns to -0.5 symbol-units per cycle and the mean symbol is 1.499992 against 1.5, the residual being only the odd symbol count — so there is no static offset for the thermal system to accumulate. What the first cycle does carry is a cold-start transient, because the convolution begins from ambient rather than from a running device. The pattern was therefore tiled 100 times as a precondition: the mean ΔT settles to within 1% of the peak-to-peak wander by cycle 2 and 0.1% by cycle 3, well inside the precondition (Appendix D). Results below are from cycle 101.

Model assumptions: (1) The MRM resonance shift is small enough that the absorbed optical power at each PAM level does not change significantly during the pattern — i.e. the power-to-temperature and temperature-to-resonance mappings are treated as linear and decoupled (no feedback from resonance drift onto absorbed power). (2) The Photonic Controller (PCON) feedback loop is not modelled. The PCON is assumed to have established the DC thermal steady state before the pattern begins; its dynamic response to pattern-dependent thermal wander is excluded. See Appendix C.7.

Cycle 101 — after 100 precondition cycles

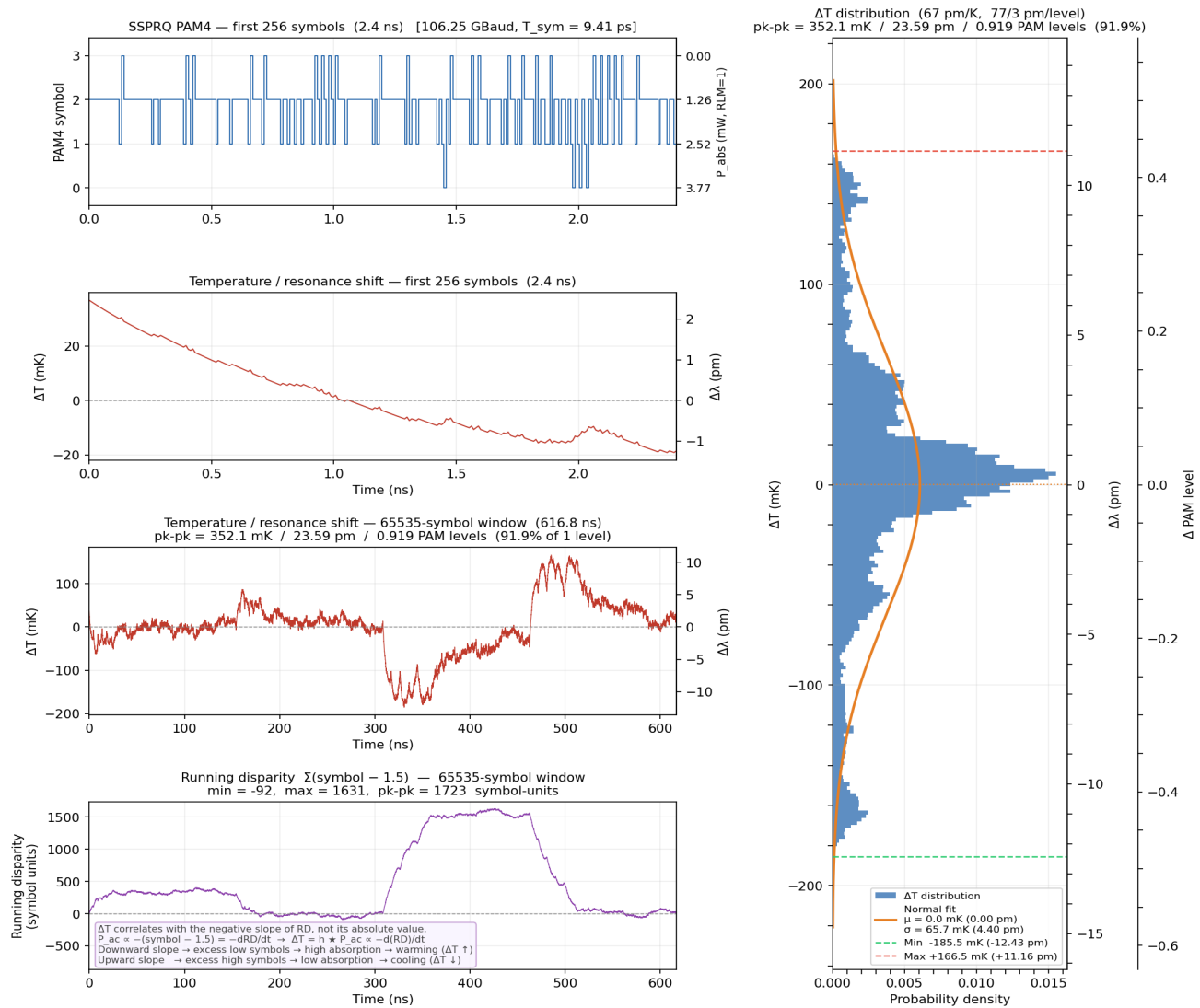


Figure 6. SSPRQ thermal wander for the Caribou-MA-DR MRM — cycle 101 (after 100 precondition cycles, periodic steady state). 65,535 symbols, 106.25 GBaud, OMA_outer = 5.767 dBm. Top-left: PAM4 symbol sequence (first 256 symbols) with absorbed-power right axis. Centre-left (zoom): delta-T for first 256 symbols with delta-lambda (pm) and delta- PAM level axes (PAM level separation = 77/3 pm). Centre-left (full): delta-T over the complete 65,535-symbol window. Bottom-left (purple): running disparity sum(symbol - 1.5) for this cycle. Right: distribution of delta-T with Gaussian fit, min/max markers, and three y-axes (mK, pm, PAM level). Peak-to-peak = 352.07 mK / 23.589 pm / 91.90% of 1 PAM level. Mean $\mu = +0.010$ mK — effectively zero; SSPRQ is DC balanced and there is no offset to accumulate (Appendix D).

4.6 Thermal Wander Results — SSPRQ Cycle 101 (Periodic Steady State)

Quantity	Value
Pattern	IEEE 802.3 SSPRQ PAM4, 65,535 symbols, 106.25 GBaud

Measurement cycle	Cycle 101 (after 100 precondition cycles; the W-fill thermal path is so fast that the mean delta-T is within 1% of converged by cycle 2)
Baud period T_sym	9.41 ps
OMA_outer	5.767 dBm = 3.773 mW
Resonance shift sensitivity	67 pm/K (Caribou-MA-DR MRM designer value)
PAM level separation	77/3 = 25.67 pm (PAM 0 to PAM 3 = 77 pm)
delta-T minimum	-185.53 mK / -12.43 pm / -48.43% of 1 level
delta-T maximum	+166.54 mK / +11.16 pm / +43.47% of 1 level
delta-T peak-to-peak	352.07 mK / 23.59 pm / 91.90% of 1 level
Normal fit — mean mu	+0.01 mK / +0.001 pm / +0.00% of 1 level (effectively zero — SSPRQ is DC balanced; Appendix D)
Normal fit — std dev sigma	65.74 mK / 4.404 pm (17.16% of 1 level)
Running disparity range (this cycle)	-92 to 1631 symbol-units (pk-pk = 1723)

Signal integrity implication: A peak-to-peak wander of 91.90% of one PAM level is a signal-integrity concern that must be considered for TDECQ and BER simulations. The Gaussian standard deviation sigma = 65.7 mK (17.16% of 1 level) characterises the RMS wander under the SSPRQ stress pattern. Unlike an imbalanced pattern, SSPRQ leaves essentially no static offset for the resonance controller to hold: the mean is +0.01 mK, 0.003% of the peak-to-peak wander. The running disparity panel shows how cumulative imbalances in the data stream build up the low-frequency thermal excursions; delta-T correlates with the negative *slope* of running disparity, not its absolute value (Appendix D).

5. Caribou-MA-OCI MRM — Full Analysis

The Caribou-MA-OCI (optical coherent interconnect) MRM is a second use case evaluated in this report. It shares the same 370 x 270 nm silicon core cross-section and W-fill contact geometry as the DR device, so the normalized thermal impulse response $h(t)$ [K m / (W s)] defined in Appendix A applies directly, with no additional FEM simulation required. The same holds for the thermal frequency response $H(f)$: it is set by the cross-section alone, so the DR and OCI curves are identical, and 5.3 reproduces that one solve rather than repeating it.

The key differences from the DR MRM are the ring radius (6.28 μm vs 9.815 μm), the modulation format (NRZ vs PAM4), and the OMA reference. Because the circumference is 39.46 μm (vs 61.67 μm for DR), the absorbed power per unit length at the OCI OMA is comparable to the DR case, giving similar temperature excursions despite the different OMA values.

5.1 OCI Simulation Parameters

Parameter	Value	Source / Notes
Ring radius R	6.280 μm	OCI device geometry
Ring circumference L	39.46 μm	2 pi R
Baud rate	106.25 GBaud	NRZ
Bit period T_bit	9.412 ps	1 / 106.25 GHz
Pattern cycle length	32,762 bits	SSPR, normative length
Pattern cycle period T_cycle	308.35 ns	comparable to tau_90 = 578.7 ns
OMA	3.793 dBm (2.395 mW)	Caribou_OMA_rollup_2026-08-31, use case 70, cell H40
NRZ OMA swing (bit 0 -> 1)	83 pm	resonance shift 0->1
PM sensitivity dLambda/dT	67 pm/K	same cross-section as DR; material / wavelength property only
Preconditioning cycles	100	ensures steady-state convergence
Pattern	SSPR (single NRZ stream)	400G CWDM8 MSA 2 km Technical Specifications, Revision 1.1, Sec 4.1.1
Pattern mean bit	0.500000000000	exactly 0.5 — blocks 5-8 invert blocks 1-4

5.2 Step-Function Thermal Response (Analytical)

Because the OCI cross-section is identical to the DR cross-section, the thermal rise times (t10, t50, t90) are exactly the same. The steady-state temperature rise scales with the absorbed power per unit length:

$$P_{\text{step_DR}} = \text{OMA}_{\text{outer}} / 2L_{\text{DR}} = 30.59 \text{ W/m}$$

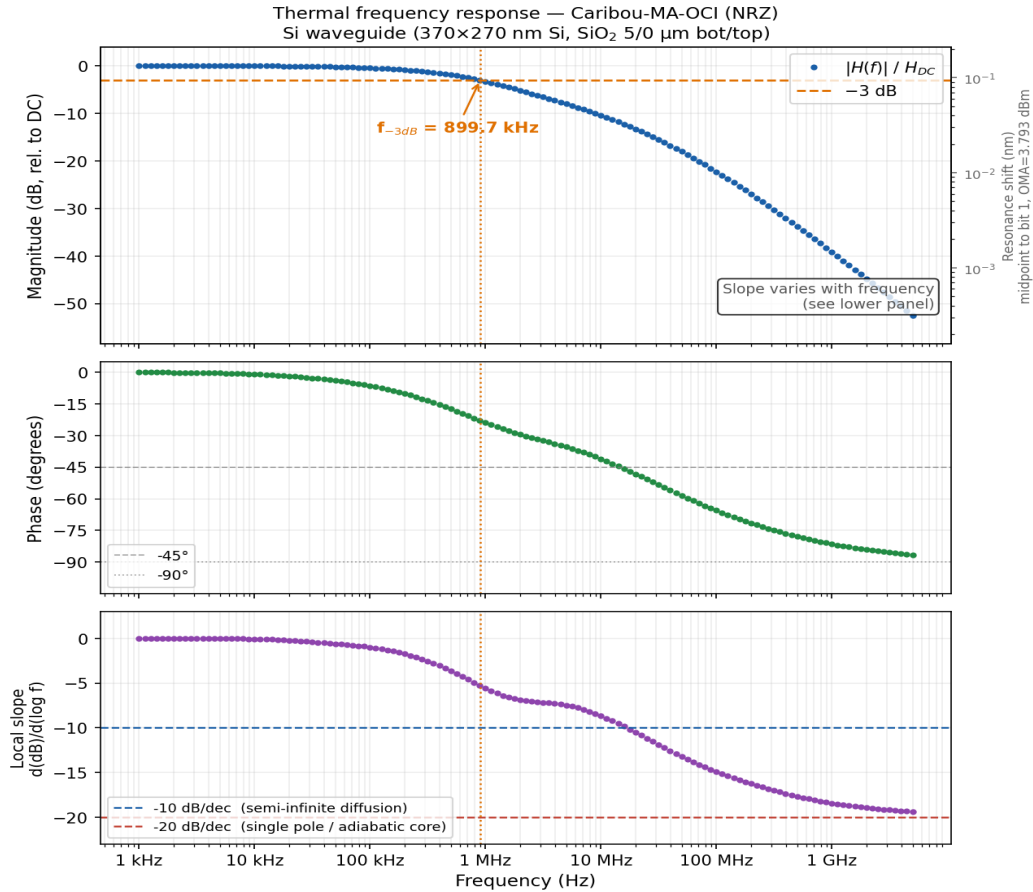
$$P_{\text{step_OCI}} = \text{OMA}_{\text{OCI}} / 2L_{\text{OCI}} = 30.35 \text{ W/m}$$

$$dT_{\text{ss_OCI}} = dT_{\text{ss_DR}} \times (P_{\text{step_OCI}} / P_{\text{step_DR}}) = 1.9996 \times 0.9920 = 1.9837 \text{ K}$$

The steady-state resonance shift is $d\Lambda_{\text{ss_OCI}} = 132.9 \text{ pm}$ (160.1% of the NRZ OMA swing). Rise times are identical to the DR case: t10 = 2.3 ns, t50 = 50.1 ns, t90 = 578.7 ns.

5.3 Thermal Frequency Response

This is the same curve as Figure 5, not a second analysis. $H(f)$ is normalised per unit absorbed power per unit length, and the OCI device has the same cross-section as the DR device, so its normalised thermal frequency response is *identical* — one phasor solve serves both. It is repeated here so this section stands on its own, with the right-hand axis rescaled to the OCI OMA/2 step; everything to the left of that axis is the shared result. 4.3 gives the full discussion, and 4.3.1 the numbers, which apply unchanged.



This is the SAME curve in both use-case sections — one phasor solve, not two. $H(f)$ is normalised per unit absorbed power and the DR and OCI devices share the cross-section exactly, so the normalised response is identical. Only the right-hand axis above, which converts it to an absolute resonance shift, is specific to this device.

Figure 7 — thermal frequency response, with the right-hand axis scaled to the OCI NRZ swing ($\Delta T_{SS} = 1.9837$ K for the OMA/2 step, against 1.9996 K for DR). The curve itself, the -3 dB frequency of 0.900 MHz and H_{dc} are the same shared quantities plotted in Figure 5; only the absolute scaling differs.

What differs between the two use cases is therefore not the response but **where each pattern puts its energy on it**. That is what makes SSPR the harsher case *in absolute terms*: it concentrates 6.5% of its swing in a single line at 3.24 MHz, close enough above the 0.90 MHz corner that a large fraction of it survives, whereas SSPRQ is close to spectrally white with its energy spread far higher (5.4.2, Figure 8).

5.4 SSPR Thermal Wander Results

The OCI stress pattern is **SSPR**, defined in 400G CWDM8 MSA 2 km Technical Specifications, Revision 1.1, Section 3.1.2. It is a single NRZ stream of 32,762 bits, applied directly to the modulator: the bitwise inverse multiplexer the MSA specifies for distributing a pattern across lanes is outside the scope of this

analysis, which characterises one MRM. SSPR is built from eight blocks: two PRBS28 sequences, a single one followed by 72 zeros, and a differentially coded copy of the first PRBS28 block, with blocks 5 to 8 the bitwise inverse of blocks 1 to 4.

The cycle period is $T_{\text{cycle}} = 308.35 \text{ ns}$, comparable to the thermal 90% rise time $\tau_{90} = 578.7 \text{ ns}$. The thermal response therefore does not settle within one pattern cycle, and preconditioning over 100 cycles is required before the wander statistics are those of the periodic steady state.

5.4.1 Wander Summary

Quantity	mK	pm	% of OMA swing
Peak-to-peak	394.91	26.459	31.88%
sigma (RMS)	96.69	6.478	7.81%
Mean offset	+0.000	+0.000	+0.00%
Minimum	-197.45	-13.229	-15.94%
Maximum	+197.45	+13.229	+15.94%

5.4.2 Exact DC Balance and Half-Wave Antisymmetry

Because blocks 5 to 8 are the bitwise inverse of blocks 1 to 4, SSPR is **exactly DC balanced**: the mean bit is 0.500000000000 and the running disparity returns to exactly zero at the end of every cycle. This is a stronger statement than the approximate balance of SSPRQ, whose net running disparity of -0 symbols per cycle drives the +0.01 mK mean offset reported for the DR case in Section 4.6. For SSPR there is no pattern-driven static thermal bias to budget at all.

The inversion has a second and stronger consequence. The absorbed power is affine in the bit, so once the controller set point (the DC mean) is subtracted, the drive satisfies $P_{\text{ac}}(t + T/2) = -P_{\text{ac}}(t)$ exactly. Convolution with any linear time-invariant response preserves that antisymmetry, so $\delta T(t + T/2) = -\delta T(t)$. The mean is therefore exactly zero and the excursion is exactly symmetric (min = -197.45 mK, max = +197.45 mK) as a structural property of the pattern, not as a numerical coincidence. Checked against the computed waveform, the residual $\max|\delta T(t) + \delta T(t + T/2)|$ is $3.80\text{e-}05 \text{ mK}$ against a 394.9 mK peak-to-peak, which is FFT round-off.

The price of exact balance is paid at low frequency. Half-wave antisymmetry removes every even harmonic, which concentrates the drive energy in the odd ones and leaves a strong fundamental at $1/T_{\text{cycle}} = 3.24 \text{ MHz}$ carrying 6.5% of the full swing amplitude. The running disparity accumulates monotonically to -615 bits over the first half of the pattern before the inverted half brings it back (range -615 to +36 bits). That excursion is a sustained excess of zeros, and in a through-port MRM a zero sits on resonance and absorbs, so it is a sustained heating interval roughly 154 ns long. At 3.24 MHz the thermal response is only 6.6 dB below its DC value (Figure 7), so about 47% of this tone's amplitude survives into the temperature.

This is why the OCI wander is the larger of the two in absolute terms — though only by about a tenth, and DR is much the worse case once each is measured against its own eye (5.4.3). Comparing the two

drive patterns on the only basis that matters thermally — how much low-frequency content they carry, weighted by the measured response:

Drive pattern property	DR (SSPRQ)	OCI (SSPR)
Running disparity, peak-to-peak	1723 symbols	650 bits
Fundamental frequency $1/T_{\text{cycle}}$	1.62 MHz	3.24 MHz
Drive amplitude at the fundamental	1.74% of swing	6.48% of swing
Whole drive spectrum after $ H(f) $ (RMS)	1.645% of swing	2.439% of swing

Each pattern is normalised to its own full drive swing, so the two columns answer “how hard does this pattern drive the thermal system, relative to its own signal amplitude”. The comparison also holds in absolute terms, because the two devices happen to absorb almost the same power per unit length (30.6 W/m for DR against 30.3 W/m for OCI) — the larger OCI ring radius offsets its smaller OMA.

SSPR presents 1.5 times as much thermally-passed drive as SSPRQ does, and its fundamental is 4 times larger. Its running disparity makes one monotonic excursion of 615 bits and back, where SSPRQ's reverses direction repeatedly and so spreads its imbalance over many frequencies. One large slow excursion is the worst case for a thermal system: it is precisely the shape the cross-section passes best.

The two drive spectra, side by side. The table above reduces each pattern to a few numbers; the figure below shows the whole spectrum of each against the measured thermal response, which is where the argument really lives. Both drives are normalised to their own full swing, and plotted as power spectral density rather than per-harmonic amplitude — the two patterns have different cycle lengths and so different harmonic spacing, and dividing by that spacing is what makes them comparable.

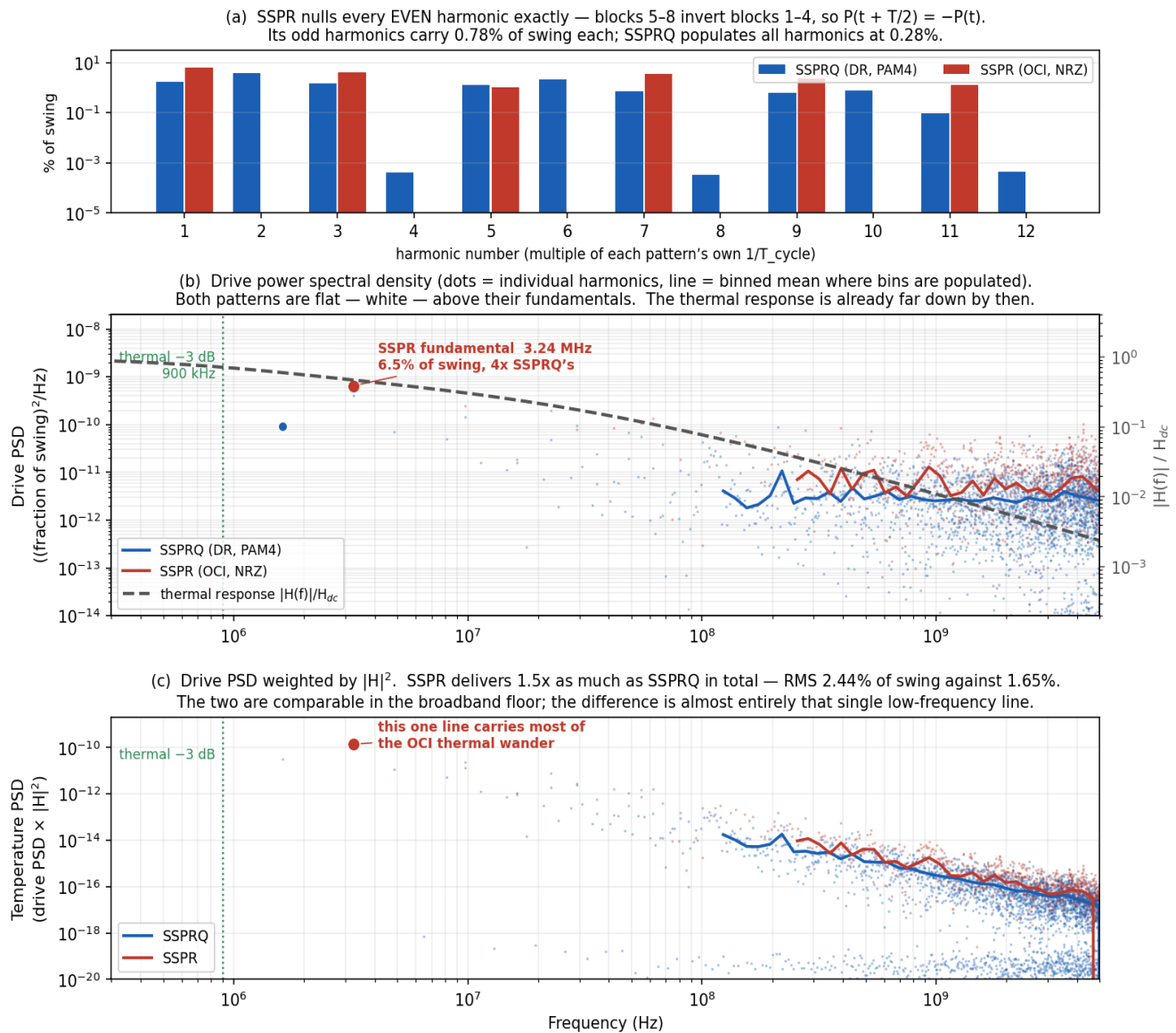


Figure 8 — drive spectra of the two test patterns against the thermal response. (a) the first twelve harmonics, where SSPR's nulled even harmonics are visible directly; (b) power spectral density over the full band, dots being individual harmonics and the line a binned mean where bins contain enough harmonics to be meaningful; (c) the same weighted by $|H(f)|^2$, which is what reaches the waveguide temperature.

Three things are worth reading off it. **First, both patterns are spectrally white above their own fundamentals** — flat density, no structure, which is what a good stress pattern should look like. On that measure the two are alike, and SSPR's broadband floor is only about 1.8 times SSPRQ's. **Second, SSPR nulls every even harmonic exactly.** That is the frequency-domain face of the half-wave antisymmetry: with blocks 5 to 8 inverting blocks 1 to 4, $P(t + T/2) = -P(t)$, and such a waveform has no even harmonics at all. The same property that guarantees exact DC balance also forces all of the pattern's power into the odd harmonics, which is why each of those carries roughly twice the density it otherwise would.

Third, both patterns carry a line at their own fundamental, and those two lines are what survive the filter. SSPR holds 6.5% of its swing at 3.24 MHz; SSPRQ holds 1.74% at 1.62 MHz, its cycle being twice as long. Panel (c) shows why that matters more than the raw amplitudes suggest. The thermal -3 dB point is at

900 kHz, so the broadband floors — where the two patterns are nearly alike — are rejected almost entirely, and what reaches the core is dominated by whatever sits closest to DC.

SSPRQ's line is the smaller of the two but sits closer to the corner, so it is attenuated less. SSPR's line is 3.7 times the larger before the filter; after it, the gap narrows to about three, and across the whole spectrum SSPR presents only 1.5 times the thermal drive of SSPRQ. The two patterns are closer in thermal severity than their spectra alone suggest, and SSPRQ is not the benign case: its measured wander is 352 mK against SSPR's 395 mK.

The practical point for pattern selection is that DC balance predicts nothing about thermal severity. Both of these patterns are DC balanced, and both present substantial thermal drive, because what the thermal system responds to is *where in the spectrum* a pattern puts its energy, not its long-term average. A pattern that balances by inverting its own second half necessarily places its largest component at half the cycle rate; a pattern with a longer cycle places its fundamental lower, and lower is worse because the thermal response is low-pass. Cycle length is therefore as much a design variable as balance.

5.4.3 Wander Figure

The figure below has five panels: (a) the SSPR bit sequence with the corresponding absorbed optical power, (b) delta-T over the first 256 bits, (c) delta-T over the full cycle, (d) the delta-T distribution with a normal fit, and (e) the running bit disparity, whose negative slope tracks the heating intervals.

SSPR NRZ Thermal Wander — Caribou-MA-OCI MRM — Cycle 101 (periodic steady state) [OMA = 3.793 dBm, 67 pm/K, 106.25 GBaud]

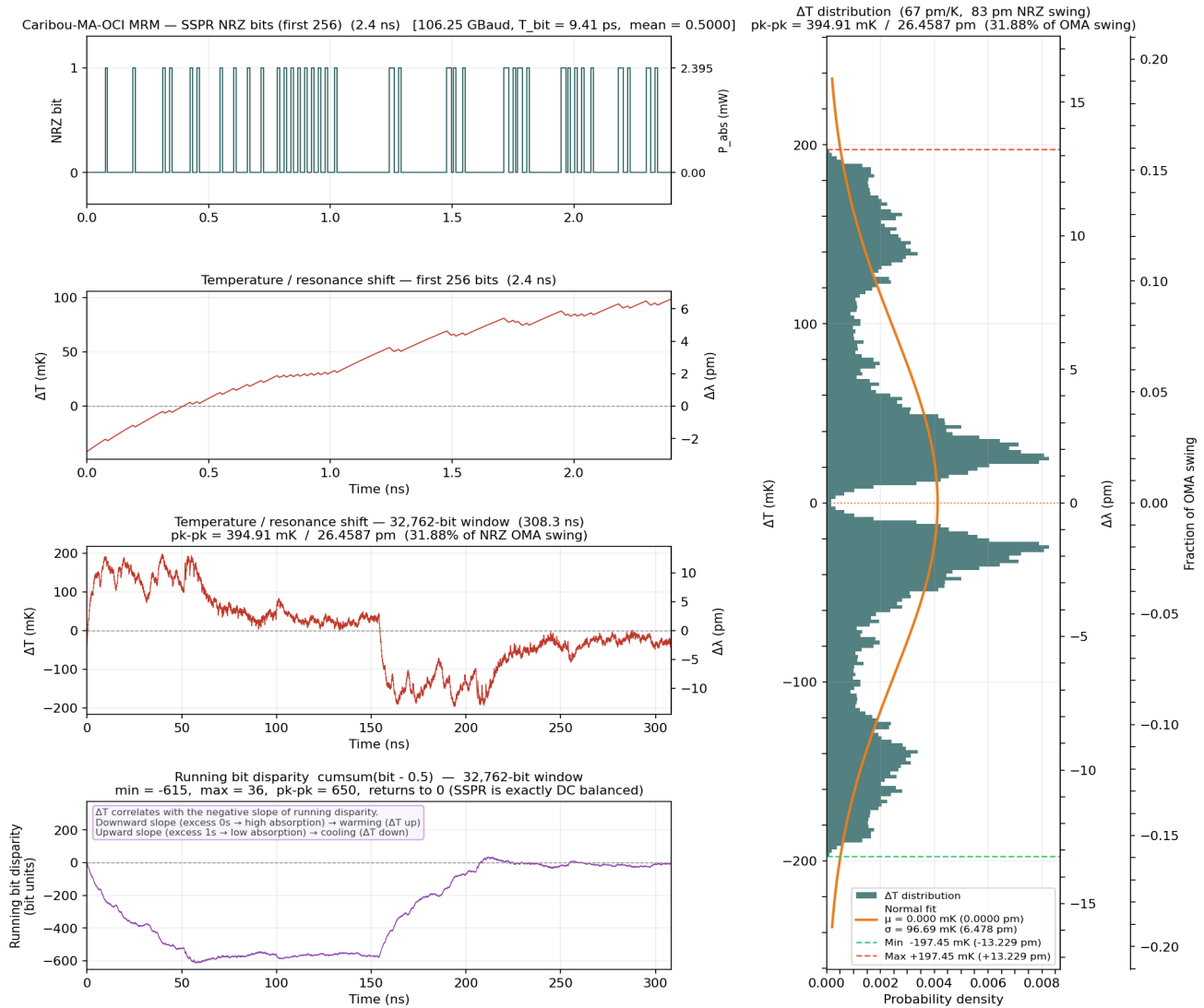


Figure 9 — SSPR thermal wander for the Caribou-MA-OCI MRM. Pk-pk = 394.91 mK = 26.459 pm = 31.88% of the 83 pm NRZ OMA swing; sigma = 96.69 mK = 6.478 pm = 7.81% of the swing; mean = +0.000 mK (exactly zero). OMA = 3.793 dBm, R = 6.28 μ m, 106.25 GBaud NRZ, cycle 101 of 101. Note the symmetry of panel (c) about zero, which is the half-wave antisymmetry of Section 5.4.2.

Signal integrity implication: the peak-to-peak wander is 26.459 pm, or 31.88% of the 83 pm NRZ OMA swing, with sigma = 96.7 mK (7.81% of the swing). NRZ signalling is more tolerant of a given absolute resonance shift than PAM4, because the full eye height is the whole OMA swing (83 pm) rather than the 25.67 pm DR PAM4 level spacing. **That tolerance is decisive, and it reverses the ranking.** The OCI excursion is 31.9% of the NRZ eye, while the DR excursion is 91.9% of a PAM level — so although OCI is larger in millikelvin, **DR is about 2.9 times worse relative to the eye it has to fit inside.** Both must be carried into TDEC and BER simulation; DR is the more constrained of the two.

6. Ideas for Future Design Improvements

This section does not describe the Caribou design. Everything else in this report models the part as built. This section models two candidate *changes* to the contact stack, so that design effort is not spent on a change that cannot pay for itself. No number in this section is a property of Caribou-MA-DR or Caribou-MA-OCI, and none of it should be quoted as one. The boundary-condition limits of C.2 apply here as everywhere else, so these are **relative comparisons between variants** computed on a common footing — not absolute predictions.

6.1 The Two Ideas, and Why They Were Tested

Two changes to the contact stack were proposed as ways to reduce the pattern-dependent self-heating.

Idea 1: raise the tungsten via fill factor. The vias are 0.36 μm on a 0.9 μm pitch, a 16% fill; doubling it to 32% means denser or wider vias over the same contact area. **Idea 2: move the tungsten closer to the core.** The W-fill onset is 2.5 μm outboard of the core edge; halving that to 1.25 μm brings the high-conductivity region nearer the heat source.

Both were carried through to the **pattern-dependent wander** rather than being judged on H_{dc} . That distinction turns out to decide the answer, so it is worth stating plainly: a change can move the steady state substantially while barely touching the wander, because raising the thermal resistance both lifts H_{dc} and lowers the roll-off corner, and above the corner those two effects largely cancel (C.2.2). Judging these ideas on H_{dc} would have given the wrong answer for both.

6.2 Results

Variant	H_{dc}	vs built	t_{90}	DR pk-pk	vs built	OCI pk-pk	vs built
	K·m/W	%		mK	%	mK	%
As built — 16%, 2.5 μm	0.0654	0.0	579 ns	352.07	0.00	394.91	0.00
Idea 1 — 32% fill	0.0643	-1.6	534 ns	352.29	+0.06	395.07	+0.04
Bound — solid W	0.0632	-3.3	490 ns	352.61	+0.15	395.35	+0.11
Bound — no W	0.0743	+13.7	1.32 μs	351.64	-0.12	394.71	-0.05
Idea 2 — W at 1.25 μm	0.0548	-16.2	285 ns	357.55	+1.56	399.77	+1.23
Ideas 1+2	0.0525	-19.7	243 ns	359.38	+2.08	401.44	+1.65

Negative means improvement. The two bound rows are not proposals: solid tungsten and no tungsten bracket every possible via density, so together they close the question rather than answering one point on it.

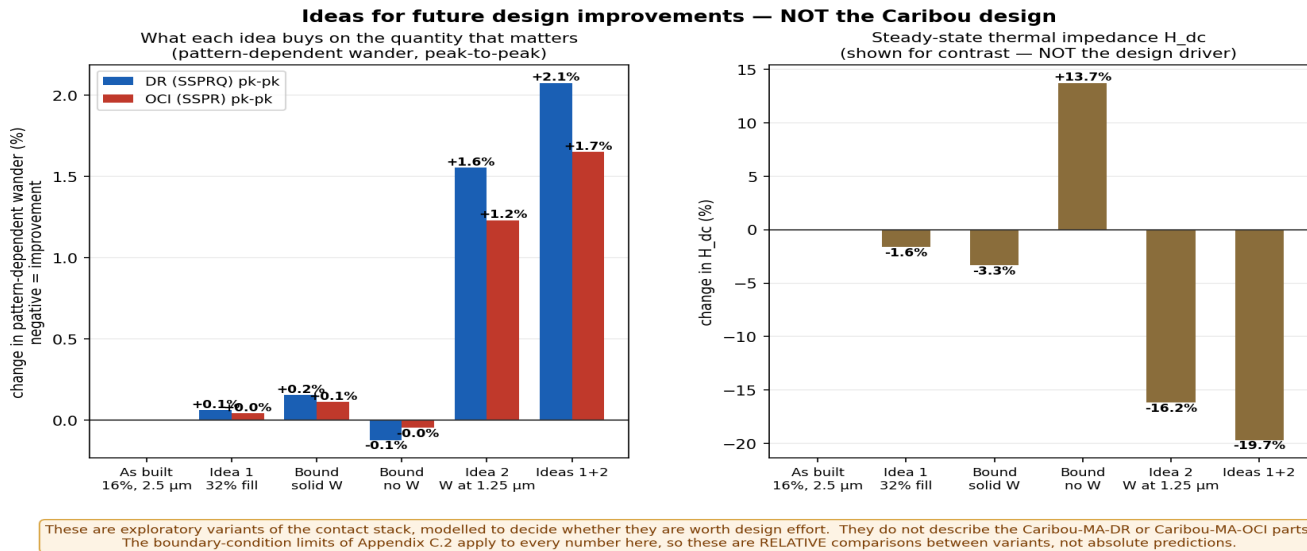


Figure 10 — what each idea buys. Left: change in pattern-dependent wander, the quantity that matters. Right: change in H_{dc} , shown only to make the point that the two are not the same measure. Exploratory variants, not the Caribou design.

6.3 Idea 1 — Via Fill Factor: No Benefit Is Available

Doubling the fill factor changes the DR wander by +0.06% and the OCI wander by +0.04%. That alone would only settle one point, so the bounding cases were run as well: a solid tungsten slab, the best any via array can possibly be, gives +0.15% and +0.11%, while removing the tungsten entirely gives -0.12% and -0.05%. Across the whole range from no tungsten to solid tungsten the DR wander moves by under 0.01% and the OCI wander by under 0.2%. **No via-density change can help, and this line of design work can be closed.**

The reason is a distinction that is easy to get backwards. About half the heat — 49% of the total — leaves through the W-fill bands, so it is tempting to conclude that the tungsten is doing half the work and that more of it must help. But **carrying heat and resisting heat are different things**. The bands are short, 0.5 μm of vertical travel, and wide, and a short wide slab of conductive material is close to a short circuit: it passes a great deal of heat while contributing very little resistance. Halving the resistance of a short circuit buys nothing. The bottleneck is upstream — getting heat laterally out of the 370 nm core through the 70 nm slab and the doped Si contacts to reach those bands — and the fill factor does not touch it.

Note how little the wander cares even when the thermal behaviour changes a great deal. Removing the tungsten raises H_{dc} by +13.7% and stretches t_{90} from 579 ns to 1.32 μs — and the wander still does not move. That is the cancellation of C.2.2 in its clearest form.

6.4 Idea 2 — Moving the Tungsten Closer: Slightly Counterproductive

This idea attacks the bottleneck that Idea 1 misses, and by the usual thermal measures it works well: halving the offset lowers H_{dc} by 16.2% and halves t_{90} , from 579 ns to 285 ns. A designer looking at those two numbers would reasonably conclude it is worth doing.

It is not. The OCI wander gets worse, by +1.23%, and doing both ideas together makes it worse still at +1.65%. DR moves the same way and slightly further (+1.56% and +2.08%), its fundamental at 1.62 MHz sitting closer to the corner than SSPR's at 3.24 MHz. Both use cases get worse, not better.

Why so little, when H_{dc} falls by 16% and t_{90} halves? It is tempting to reach for the roll-off corner: lowering the thermal resistance raises the corner, moving it toward the tone SSPR concentrates its energy in, which would let more of that tone through. **That explanation does not survive contact with these numbers.** Taken literally — wander proportional to H_{dc} times the corner frequency — it predicts +70% for Idea 2, +91% for the two together, and -50% for removing the tungsten. The measured values are +1.23%, +1.65% and -0.05%. The direction happens to come out right in all three, but the size is wrong by factors of 55 to 1000, which leaves the agreement in sign worth nothing. The response is not single-pole (4.3.1), and a single-pole intuition is not a safe guide to it.

The explanation that does hold is simpler, and it is the same one that governs 6.3. **Every variant in this section changes the thermal path in the far field** — the contact stack sits 1.25 to 2.5 μm out from the core edge — whereas the pattern-dependent wander is set by the near field, the material immediately around the core and slab. Far-field changes move H_{dc} and t_{90} a great deal and the wander almost not at all.

What separates the two use cases is which frequencies each pattern occupies. SSPR puts 6.5% of its swing into a single line at 3.24 MHz (5.4.2), a 308 ns period, which sits close to the edge of the window over which this model is near-field dominated (about 190 ns on a 1% criterion, C.2.1). That is why OCI shows a small but non-zero sensitivity to far-field changes while DR — close to spectrally white, with its energy far higher (5.4.2, Figure 8) — shows essentially none. The residual is small enough, and the simple corner argument unreliable enough, that its *sign* should not be over-interpreted; what matters for planning is the magnitude, which is at most 2.1% across every variant and both use cases.

6.5 Conclusion for Design Planning

Neither idea is worth design time. The fill factor cannot help at any density, and the bounding cases prove it rather than suggesting it. Moving the tungsten closer improves every conventional thermal metric while making *both* use cases slightly worse — DR by rather more than OCI — so it should not be adopted for this purpose either.

The general lesson is that the contact stack is not where this problem lives. Every variant tested here changes the thermal path *outside* the waveguide, and the pattern-dependent wander is governed by the near field and by the drive spectrum, neither of which those changes reach. A change that genuinely reduces this wander has to either alter the near-field geometry of the core and slab, or alter what the pattern puts into the thermal passband. Both patterns place a line near the corner — SSPR at 3.24 MHz and SSPRQ, with its longer cycle, at 1.62 MHz — and moving or breaking up those lines is likely to be worth more than any amount of tungsten. Cycle length is the lever, and it is a system choice rather than a process one.

The contrast with the undercut shows where the leverage actually is. Removing the lateral silicon alters the structure immediately beside the core and slab — a near-field change — and it moves the OCI wander by +26% (C.2.2), against at most 2.1% for every contact-stack variant tested here. Proximity to the core, not conductivity or thermal resistance, is what decides whether a change reaches this problem.

Appendix A — Mathematical Formulation

A.1 Governing Equation

The 2D transient heat equation in the waveguide cross-section (x, y) is:

$$\rho \cdot C_p(x,y) \cdot dT/dt = \text{div}[k(x,y) \cdot \text{grad}(T)] + Q(x,y)$$

where $T(x,y,t)$ is the temperature rise above ambient, k is thermal conductivity, ρ is density, C_p is specific heat, and Q is the volumetric heat source (W/m^3). The material properties are piecewise constant: Si core/slabs/contacts, SiO_2 cladding, and W-fill composite (16% W via-fill + 84% SiO_2) above the Si contacts from $2.5 \mu\text{m}$ outboard of the core edge out to the lateral domain boundary. The upper boundary ($0.5 \mu\text{m}$ above the core) and lower boundary ($5 \mu\text{m}$ below the core) are held at $T = 0$ (perfect heatsink). The scalar output used throughout — the step response $S(t)$, the thermal wander $\Delta T(t)$, and all figures and tables — is the **spatial mean of T over the 370x270 nm Si core**: $T_{\text{core}}(t) = (1/A) [\text{area integral over Si core}] T(x,y,t) dx dy$, where $A = 370 \text{ nm} \times 270 \text{ nm}$. This is justified because the optical mode is confined within the core and k_{Si} is large ($85 \text{ W}/(\text{m}\cdot\text{K})$), so intra-core temperature gradients are negligible on the timescales of interest.

A.2 Spatial Discretisation

The domain is discretised on a uniform 10 nm Cartesian grid. The full cross-section is $N_x = 1,038$ by $N_y = 578 = 599,964$ nodes. The grid pitch is chosen so that every material boundary falls exactly on a node, so no feature is rounded to suit the mesh. Because the cross-section is mirror-symmetric about the core centre, only the left half is solved -- $519 \times 578 = 299,982$ nodes -- with a zero-flux symmetry plane on the cut face. That reduction was validated against the full domain: dT_{ss} , H_{dc} and the entire step response agree to $7\text{e-}08$ relative. The term $\text{div}(k \cdot \text{grad } T)$ is approximated with a five-point finite-difference stencil. At material interfaces the interface conductivity is taken as the harmonic mean of the adjacent cell conductivities:

$$k_{i+1/2} = 2 k_i k_{i+1} / (k_i + k_{i+1})$$

This is exact for a 1D two-layer problem and significantly improves accuracy at the Si/SiO_2 interface where k differs by a factor of 94. Dirichlet boundary conditions ($T = 0$) are enforced implicitly: the boundary rows of the spatial operator L are set to zero, so the boundary temperatures remain zero throughout the time integration without explicit projection.

Assembling the stencil coefficients into a sparse $N \times N$ matrix L , the semi-discrete system is:

$$\text{diag}(\rho \cdot C_p) \cdot dT/dt = L \cdot T + Q$$

A.3 Time Integration — Decade-Blocked Backward Euler

The full time range (1 ps to 20 ms) spans 11 decade blocks. A uniform log-spaced grid would require a large dt at late times (wasting solves) or a tiny dt at early times (excessive cost). Instead, the range is divided into 11 decade blocks; within each block a fixed dt is used, giving 200 uniform steps per block (2200 total). Each step applies the backward-Euler (implicit) scheme:

$$[\rho \cdot C_p / dt \cdot I - L] \cdot T_{n+1} = \rho \cdot C_p / dt \cdot T_n + Q$$

Because dt is constant within a block, the left-hand-side matrix is constant and its sparse LU factorisation (SuperLU) is computed once per block and reused for all 200 solves via forward/back substitution. This gives only 11 expensive factorisations for 2200 time steps. Backward Euler is unconditionally stable (no time-step stability constraint).

The finest step size is $dt = 45$ fs in the 1-10 ps decade. Each step is a real sparse solve with 299,982 unknowns.

Appendix B — Verification of Mathematical Approaches

Each major modelling choice was independently verified. This appendix records those checks.

B.1 Harmonic Mean Interface Conductivity

At the Si/SiO₂ interface, the interface conductivity is $k_{i+1/2} = 2k_i k_{i+1} / (k_i + k_{i+1})$. This is exact for a 1D two-layer problem on a uniform grid: placing two half-cell resistances $dx/(2k_i)$ and $dx/(2k_{i+1})$ in series and solving for the effective single-cell resistance dx/k_{eff} gives exactly the harmonic mean. With $k_{\text{Si}}/k_{\text{SiO}_2} = 94$, the choice of interface averaging matters substantially; arithmetic or geometric means would introduce O(1) errors in the interfacial heat flux. The harmonic mean eliminates this.

B.2 Dirichlet Boundary Condition Enforcement

Boundary nodes are excluded from the finite-difference stencil, so their rows in L are identically zero. The backward-Euler system at a boundary node reduces to $(\rho \cdot \text{Cp}/dt) \cdot T_{n+1} = (\rho \cdot \text{Cp}/dt) \cdot T_n$, which gives $T_{n+1} = T_n$. Since the initial condition $T_0 = 0$ everywhere, the boundary values remain 0 at every subsequent step — $T = 0$ is enforced for all time without any explicit projection step. In the frequency domain the same rows give $(j \cdot \omega \cdot \rho \cdot \text{Cp}) \cdot T_{\text{hat}} = 0$; since $j \cdot \omega \cdot \rho \cdot \text{Cp}$ is non-zero for $f > 0$, $T_{\text{hat}} = 0$ at boundaries for all frequencies in the sweep.

B.3 Fourier Integral Identity

The frequency response $H(f)$ equals the Fourier transform of the impulse response $h(t) = dS/dt$. Integrating by parts:

$$H(f) = \int_0^\infty (dS/dt) \cdot \exp(-j \cdot \omega \cdot t) dt = [S(t) \cdot \exp(-j \cdot \omega \cdot t)]_0^\infty + j \cdot \omega \cdot \int_0^\infty S(t) \cdot \exp(-j \cdot \omega \cdot t) dt$$

The boundary term vanishes: $S(0)=0$ (zero initial temperature) and $S(t) \cdot \exp(-j \cdot \omega \cdot t) \rightarrow 0$ as $t \rightarrow \infty$ for $\omega > 0$. Splitting $S(t) = [S(t) - H_{\text{DC}}] + H_{\text{DC}}$ and noting that $j \cdot \omega \cdot \int_0^\infty \exp(-j \cdot \omega \cdot t) dt = 1$ for $\omega > 0$ gives the identity $H(f) = H_{\text{DC}} + j \cdot \omega \cdot \int_0^\infty [S(t) - H_{\text{DC}}] \cdot \exp(-j \cdot \omega \cdot t) dt$. The subtraction of H_{DC} is necessary for convergence: $S(t) - H_{\text{DC}} \rightarrow 0$ as $t \rightarrow \infty$, so the integrand decays; without it the integral is conditionally convergent and numerically unstable.

B.4 Piecewise-Linear Segment Integrals

For segment $[t_k, t_{k+1}]$ with S linear in $\tau = t - t_k$ from S_k to S_{k+1} , letting $a_k = S_k - H_{\text{DC}}$ and slope $k = (S_{k+1} - S_k) / dt_k$:

$$\int_0^{dt_k} (a_k + \text{slope}_k \cdot \tau) \cdot \exp(-j \cdot \omega \cdot \tau) d(\tau) \\ = a_k \cdot I_1 + \text{slope}_k \cdot I_2$$

where $I_1 = \int_0^{dt_k} \exp(-j \cdot \omega \cdot \tau) d(\tau) = (1 - \exp(-j \cdot \omega \cdot dt_k)) / (j \cdot \omega)$ and $I_2 = \int_0^{dt_k} \tau \cdot \exp(-j \cdot \omega \cdot \tau) d(\tau) = -dt_k \cdot \exp(-j \cdot \omega \cdot dt_k) / (j \cdot \omega) + I_1 / (j \cdot \omega)$, the latter obtained by integration by parts. Multiplying by the phase factor $\exp(-j \cdot \omega \cdot t_k)$ shifts the integral from the local coordinate τ back to the global time t . Both integrals are analytic and carry no sampling or Nyquist constraint.

B.5 Time-Domain / Frequency-Domain Cross-Check

As a final verification, the thermal frequency response $H(f)$ computed by the direct phasor solve is compared against the Fourier transform of the time-domain step response $S(t)$ loaded from the FEM simulation. The two routes are fully independent: the phasor solve never uses $S(t)$, and the Fourier integral never uses the phasor matrices. Since the cross-section geometry is identical for both DR and OCI, one FD phasor solve and one Fourier integral validate the normalised $H(f)/H_{dc}$ for both devices. H_{dc} is itself geometric, so there is no separate per-device DC gain to cross-check -- only a single shared curve.

Figure B.5 overlays the two routes. The error panel (bottom) shows the difference in dB; agreement across the whole 1 kHz to 5 GHz sweep confirms that the impulse response $h(t)$ stored in the CSV accurately represents the FEM system's transfer function, to 0.027 dB RMS and 0.053 dB at worst.

Time-domain / frequency-domain cross-check
Shared cross-section (Caribou-MA-DR and Caribou-MA-OCI) | $H_{dc} = 0.065364$ K.m/W | RMS error = 0.027 dB

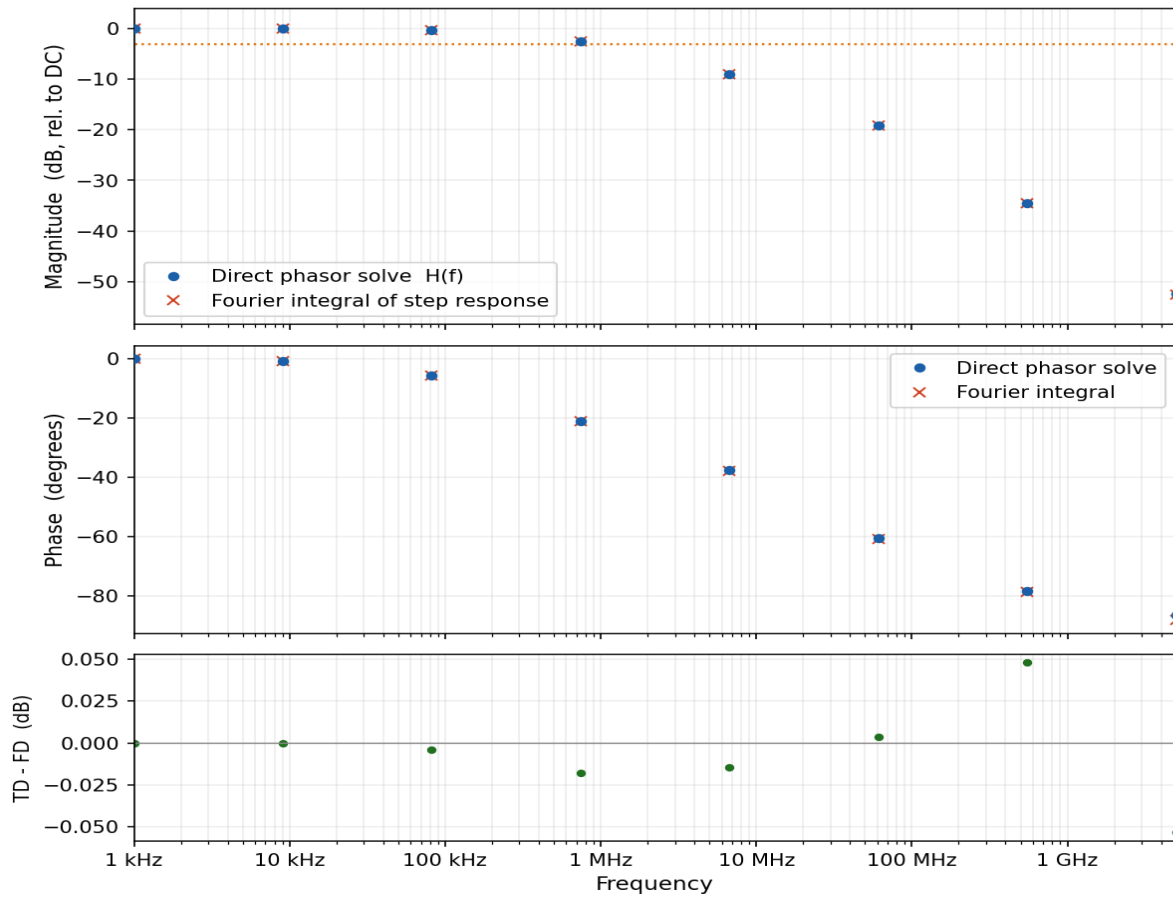


Figure B.5 — time-domain Fourier integral (red dashed) vs direct phasor solve (blue solid). Magnitude (top), phase (centre), error in dB (bottom). One figure serves both use cases: $H(f)$ is normalised per unit absorbed power, so it — and H_{dc} with it — is a property of the shared cross-section, identical for DR and OCI. What differs between the devices is only the absolute rise,

$$\Delta T_{ss} = H_{dc} \times P_{step}$$

Appendix C — Modeling Assumptions and Caveats

The following assumptions are built into the model. Each is appropriate for a first-pass analysis but should be revisited before making quantitative design decisions. **C.1 is the governing one** — it sets what question this report can and cannot answer, and the rest are refinements by comparison.

C.1 Heat Source Computed with Self-Heating Ignored — the Governing Approximation

The time-dependent heat source in this model is computed from the MRM's output OMA **with this pattern-dependent self-heating ignored**. Specifically, the heat source is the difference in MRM output power between PAM 3 and PAM 0 for the PAM4 scenarios, or between level 1 and level 0 for the NRZ scenarios — that is, the OMA — again with this pattern-dependent self-heating ignored.

Only that difference enters the calculation. Light that does not leave the through port is absorbed in the ring — apart from scattering and any drop-port leakage, treated separately in C.9 — so a swing in output power corresponds to an equal and opposite swing in absorbed power. The model spreads that swing across the equally spaced levels and then subtracts the mean, because a constant absorbed power would add only a static temperature rise, which the controller holds (C.8). The absolute absorption at any one level therefore never has to be known, and no assumption is made about where the resonance sits relative to the laser line.

The real device is a closed loop. Absorbed power raises the core temperature; temperature red-shifts the resonance at 67 pm/K; a shifted resonance changes how much of the incident light the ring absorbs at each drive level; and that changes the absorbed power:

$$Q \rightarrow \Delta T \rightarrow \Delta \lambda \rightarrow \text{absorption} \rightarrow Q$$

This model cuts that loop open. It assumes the modulator is delivering its design OMA and absorbing the corresponding power — that is, it **deliberately assumes the self-heating being calculated has not itself degraded the modulator**. The degradation in OMA caused by pattern-dependent self-heating is not modelled.

Why the approximation is deliberate, and what question it answers.

The aim is not to predict the degraded OMA with high accuracy. It is to establish whether pattern-dependent self-heating is a **significant** effect at all, and for that purpose the two possible outcomes are asymmetric:

- **If the computed wander is a small fraction of the eye**, then the resonance excursion is small compared with the modulation swing, so the perturbation to absorption is small, so the error in Q is small. The correction to the wander is second order in the wander itself. The approximation is then self-consistent, and the verdict — that self-heating can be neglected — stands without any further modelling.
- **If the computed wander is a large fraction of the eye**, the approximation is quantitatively wrong, but the verdict is already that the effect is significant. A more exact figure would not change the engineering consequence, which is that the optical link cannot be budgeted as though self-heating were absent.

The approximation therefore cannot manufacture a false negative. A spurious “insignificant” verdict would require the neglected feedback to be large while the wander is small, and the feedback is *driven by* the wander — so that combination is self-contradictory. This is what makes a deliberately optimistic heat source

the right choice for a screening calculation.

What the results say.

The wander is not small. Expressed as the resonance excursion it drives, against the modulation swing it has to be compared with:

Quantity	Caribou-MA-DR (SSPRQ)	Caribou-MA-OCI (SSPR)
Modulation swing	25.67 pm (1 PAM level)	83 pm (full OMA swing)
Wander, peak-to-peak	23.59 pm	26.46 pm
as a fraction of the swing	91.9%	31.9%
Uncompensated DC shift	134 pm (522% of a level)	133 pm (160% of the swing)

Two things follow. First, the **DC** self-heating is larger than the eye itself, so the device depends on its controller to hold the operating point at all; that is a design premise rather than a finding, and the controller is discussed in C.8. Second, the **residual pattern-dependent** excursion that the controller cannot track is still 92% of a PAM level for DR and 32% of the OMA swing for OCI. Those are not small perturbations to the absorption, so the significance test has returned **significant** — which is the answer it was built to give.

This approximation is consequently the limiting one for any quantitative use of the absolute amplitudes in this report. Turning these figures into design numbers requires a self-consistent calculation, in which the absorption at each drive level is evaluated at the instantaneous resonance position rather than the nominal one.

One caveat on the sign of the neglected feedback.

Whether the loop is self-limiting or regenerative depends on which side of the resonance the operating point sits. If heating moves the resonance away from the laser line, absorption falls and the excursion is damped. If heating moves it toward the laser line, absorption rises and the excursion grows — the thermal bistability familiar from silicon microrings. The magnitude argument above holds either way while the wander is small. But in the regime the results actually show, a regenerative sign would make the true excursions **larger** than computed, not smaller. The amplitudes here are therefore better read as a lower bound on the concern than as an estimate of it.

What this approximation does not affect.

The linear dynamics are properties of the cross-section, not of the drive. The impulse response $h(t)$, the frequency response $H(f)$, the rise times, the thermal memory and the shape of the roll-off all follow from the geometry and the material properties, and are independent of the OMA the ring is asked to deliver. The conclusions about **timescales**, about the roll-off not being single-pole (4.3.1), and about the **relative** severity of different test patterns (4.5, 5.4) are therefore unaffected by the fixed-Q assumption. It is the absolute wander amplitudes, and only those, that carry this uncertainty.

C.2 Thermal Boundary Conditions — and the Timescale They Are Valid Over

This is the approximation that decides which results in this report are predictive and which are not.

The domain is 10.37 μm wide by 5.77 μm tall, with Dirichlet $T = 0$ at top and bottom and Neumann $dT/dx = 0$ at the sides. Those boundaries were placed to resolve the near field of the waveguide faithfully, because that is what governs the pattern-dependent wander this report is about. **They deliberately do not represent the real structure at those distances**, and beyond a certain time it is the boundaries rather than the device that set the answer.

C.2.1 When the Boundaries Begin to Matter

Two questions have to be separated. The first is when heat *reaches* a boundary. For diffusion from a suddenly heated surface, $T/T_{\text{surface}} = \text{erfc}(x/\sqrt{4\cdot\alpha\cdot t})$, so requiring the disturbance at distance L to have reached 1% of the driving rise fixes the time as $L^2/(13.3\cdot\alpha)$. (The familiar penetration depth $\sqrt{4\cdot\alpha\cdot t}$ is the $\text{erfc}(1) = 15.7\%$ contour, which is 3.3 times later; a criterion has to be stated or the number means nothing.)

Boundary or feature	Distance	Medium	Heat arrives (1% contour)
Top Dirichlet sink	0.5 μm above core	SiO_2	25 ns
Lateral Si contacts	4.61 μm	Si	35 ns
W-fill onset	2.5 μm outboard	SiO_2	621 ns
Bottom Dirichlet sink	5 μm below core	SiO_2	2,486 ns

Three things about that table are worth noticing. First, the top sink is reached *sooner* than the far lateral silicon even though the oxide is 60 times less diffusive, because the contacts are 9.2 times further away and distance enters squared. 0.5 μm of oxide is thermally about as far as 3.9 μm of silicon, and the contacts run slightly further than that.

Second, the two Dirichlet sinks are 100 times apart in time even though they are the same material — the bottom is 10 times further away and the time goes as the square of that. The three arrival times therefore tell one story: the top sink and the lateral silicon engage together within a few tens of nanoseconds and between them govern everything this report measures, while the bottom sink takes no part in the transient at all and shows up only in the steady state. That is consistent with where the heat actually ends up — 94.9% through the top against 5.1% through the bottom (C.2.5), a ratio of 19:1 where plain oxide alone would give about 10:1; the excess is the W-fill raising the top path to 25.93 against 1.53 W/(m·K).

Third, these are one-dimensional semi-infinite estimates used in a two-dimensional structure, so they are order-of-magnitude guides and not all equally trustworthy. The top row is the most defensible: 0.5 μm of oxide sitting directly above the core is close to the one-dimensional case the formula describes. The lateral row is the weakest, because it treats a 270 nm silicon layer embedded in oxide, reached through a 70 nm slab, as though it were bulk silicon; heat leaks into the surrounding oxide all along that path, so the real lateral transit is slower than the table suggests. None of this changes the conclusion, because the conclusion rests on the measured divergence below rather than on these estimates.

The second question is the one that matters: when does *where the boundary was put* change the answer? That is later than arrival, because the core does not care until the boundary begins removing a meaningful

share of the heat and the effect propagates back. It is also measurable rather than estimable, so it was measured: the transient was solved twice with the top cladding at 0.5 μm and at 1.0 μm , and the first time the two core temperatures diverge was recorded (boundary_sensitivity.py).

Core temperature differs by more than	beyond
0.5%	150 ns
1%	190 ns
2%	248 ns
5%	406 ns
10%	748 ns
20%	never

So on a 1% criterion this model is trustworthy out to roughly 190 ns, and progressively less so after that. The step response passes 50% of its final value at 50.1 ns, inside that window, so about half the transient is genuinely insensitive to the boundary. By 578.7 ns (t_{90}) the divergence is already 5% and rising, and at the end of the run a mere doubling of the cladding has changed the steady state by 18% (0.9998 K against 1.1803 K). The real structure differs from this model by far more than a factor of two in that dimension, which is why the steady-state error is a factor of six rather than a fifth.

Note that the measured 190 ns is later than the 25 ns arrival estimate in the table above, as expected: arrival is necessary for the boundary to matter but not sufficient. The wander correlation time is about 134 ns and SSPR's dominant tone has a 308 ns period, so the pattern-dependent behaviour sits around and just beyond this limit — which is why its sensitivity was tested directly rather than assumed, below.

C.2.2 Consequence: the Steady-State Numbers Are Not Predictions

The real part is undercut — the silicon beneath and beside the ring is removed to isolate it thermally — and this model has no such feature. It instead provides an intact lateral silicon path and a perfect sink 0.5 μm above the core, both of which carry heat away far too readily. The result is a steady-state thermal resistance of 1.06 K/mW for the DR ring against roughly 6.2 K/mW measured by the foundry — low by about a factor of six. The foundry figure is the authoritative one for steady state; there is good measured data for it, and nothing in this model should be used in its place.

Every quantity that depends on the far field inherits that error and must not be quoted as a device property: the steady-state rise $\Delta T_{ss} = 1.9996$ K, the 134.0 pm static resonance shift, the DC thermal impedance H_{dc} , the ring thermal resistance, and the rise times beyond roughly 25 ns. They appear in this report because the model needs an internally consistent normalisation — $h(t)$ is defined against ΔT_{ss} , and the frequency response is referenced to H_{dc} — not because they describe the part.

C.2.3 What Survives

The pattern-dependent wander is far less exposed than the raw size of the error suggests, because the two effects of a more isolated ring act against each other: a higher thermal resistance raises H_{dc} , but the same change lengthens the time constant and moves the roll-off corner down by very nearly the same factor, so

the response at the frequencies a pattern actually occupies barely moves. Tested directly by removing the lateral silicon path, a 2.28x rise in steady-state resistance changed the OCI wander by +26%. *That test predates the SSPRQ correction of F.2, so no DR figure from it is quoted; the OCI figure stands because SSPR did not change. Re-measuring the DR sensitivity to the undercut is outstanding, and on the contact-stack variants of 6.4 DR is the more sensitive of the two, so it should not be assumed smaller.* The conclusions about relative pattern severity, about the shape of the roll-off, and about the non-Gaussian distributions are properties of the near field and of the drive spectra, and stand. Both patterns place a line near the corner — SSPR at 3.24 MHz and SSPRQ, with its longer cycle, at 1.62 MHz — so both are exposed to further isolation moving the corner toward them (5.4.2). On the variants tested in 6.4 the DR figure moves slightly more than the OCI one, its line being the closer of the two to the corner.

C.2.4 What a Faithful Model Would Require, and Why It Is Not Attempted Here

Representing the structure faithfully at these distances means the undercut, the full metal stack above, and the route to the substrate — none of which is a two-dimensional cross-section. Heat spreads along the ring as well as across it, and the undercut geometry is not uniform around the circumference, so an honest treatment is a three-dimensional transient model. Those are notoriously difficult to get right: meshing, contact conductances and convergence all bite, and errors in them are quiet rather than obvious. That work belongs in a commercial simulation tool with deep engineering behind it, validated against the foundry's measurements, and not in a model assembled quickly for a specific question. The sensible division is to take steady-state behaviour from the foundry's measured data, and to use this model for what a careful two-dimensional near-field treatment can legitimately provide: the sub-microsecond, pattern-dependent dynamics that the measured DC figure says nothing about.

C.2.5 Where the Heat Goes in This Model

Boundary	Condition	Distance from core	Share of steady-state heat
Top	$T = 0$ (heatsink)	0.5 μm above core top	94.9 %
Bottom	$T = 0$ (heatsink)	5.0 μm below core	5.1 %
Left	$dT/dx = 0$ (insulating)	5.19 μm from centre	0 % (by construction)
Right	$dT/dx = 0$ (insulating)	5.19 μm from centre	0 % (by construction)

The top boundary carries 94.9% of it, of which 52% leaves through the two W-fill bands and 48% through the SiO₂ window directly above the core. The bottom carries only 5.1%, being 10x further away through a medium whose conductivity is 17x lower than the W-fill composite. The lateral condition forbids sideways spreading, which is conservative in isolation; at steady state the hottest lateral node reaches 0.116 K against 2.028 K in the core, 5.7% of peak, so the domain is not quite wide enough for that condition to be free. All of this describes the model, not the part.

C.3 10 nm Grid — Resolution at Interface Gradient

The waveguide (370×270 nm) is resolved by $37 \times 27 = 999$ cells on the 10 nm grid, an exact fit in both directions. At the Si/SiO₂ interface, the temperature gradient in SiO₂ is 56× steeper than in Si (to satisfy flux continuity), so a single 10 nm cell outside the waveguide must carry this gradient. The harmonic mean interface conductivity ensures the flux is correct, but the spatial temperature profile near the interface has limited resolution. A grid refinement study (halving to 5 nm, quadrupling the node count) would quantify the residual error in mean waveguide temperature.

C.4 Material Properties — Values, Temperature Dependence and Sources

The properties are evaluated at the operating temperature, not at room temperature. The device runs between 105 °C and 260 °C, and this model is built at $T_{op} = 150$ °C (423.15 K). That choice is not cosmetic. Between 300 K and 423 K the thermal conductivity of silicon falls by about a third and the heat capacity of silica rises by about a quarter; together they raise the steady-state temperature rise by roughly 40% and lengthen every thermal time constant by about two thirds. A room-temperature property set would not be a small error here.

Each property is implemented in *device_params.py* as a function of temperature, fitted to or taken directly from the primary source cited below; the values used are those functions evaluated at T_{op} . The operating temperature is a single configurable constant, so the whole property set and every result downstream follow from it. **The cited papers are in the project Google Drive folder alongside the published report.**

C.4.1 Values at $T_{op} = 150$ °C

Property	Value	Source
k, Si core and slab	85.3 W/(m·K)	Ohishi 2015, Table IV (see C.4.2)
k, Si contacts (doped)	65.5 W/(m·K)	Ohishi 2015, Table IV (see C.4.3)
Cp, Si	800.3 J/(kg·K)	NIST-JANAF, Shomate for Si(cr)
rho, Si	2,330 kg/m ³	handbook, 300 K
k, SiO ₂	1.53 W/(m·K)	handbook 300 K + Wray & Connolly slope
Cp, SiO ₂	916.2 J/(kg·K)	NIST-JANAF, Shomate for alpha-quartz
rho, SiO ₂	2,200 kg/m ³	handbook, 300 K
k, W	154.0 W/(m·K)	Tolias 2017, Hust-Lankford fit
Cp, W	137.0 J/(kg·K)	handbook
rho, W	19,250 kg/m ³	handbook, 300 K
k, W-fill composite	25.93 W/(m·K)	16% W by volume, parallel path
rho·Cp, W-fill composite	2,115,122 J/(m ³ ·K)	volume average of rho·Cp (see C.4.5)

C.4.2 Silicon Thermal Conductivity

Silicon thermal conductivity is taken from Y. Ohishi, J. Xie, Y. Miyazaki, Y. Aikebaier, H. Muta, K. Kurosaki, S. Yamanaka, N. Uchida and T. Tada, *Thermoelectric properties of heavily boron- and phosphorus-doped silicon*, Japanese Journal of Applied Physics **54**, 071301 (2015), Table IV. That work measures single-crystal Si over 10^{18} – 10^{20} cm⁻³ for both dopants, with conductivity derived from laser-flash diffusivity together with heat capacity and density. It suits this model for three reasons: it spans exactly the doping range in question, it reports both dopant species, and it covers room temperature to above 1000 K so that T_{op} is reached by interpolation rather than extrapolation.

The paper also confirms that the electronic contribution is negligible here: computed from the measured electrical conductivity through the Wiedemann–Franz law, it is below 0.8 W/(m·K) in every sample. The conductivity used is therefore lattice conduction, and the same phonon physics governs both the doping and the temperature dependence.

Table IV reports kappa at 300 K and 570 K for six samples:

Sample	Dopant	Carrier density (cm ⁻³)	kappa, 300 K	kappa, 570 K
N3	P	1.1×10^{18}	127	62
P3	B	1.8×10^{18}	124	60
P2	B	6.3×10^{18}	122	62
N2	P	6.7×10^{18}	113	57
N1	P	6.8×10^{19}	85	48
P1	B	1.2×10^{20}	73	48

Two interpolations take this to an arbitrary doping and temperature, in the order that keeps each one inside measured data. First, in log(doping) between the bracketing samples, performed separately at 300 K and at 570 K, since kappa varies close to linearly with log N in this regime. Second, in temperature, as a power law kappa proportional to T^{-n} with the exponent taken from that same interpolated pair. Fitting the exponent per doping level matters: it falls from about 1.12 when lightly doped to about 0.65 at 10^{20} cm⁻³, because impurity scattering loses ground to Umklapp scattering as the impurity density rises. The implementation is asserted to reproduce all twelve Table IV entries exactly at their own measured points.

The core and the shoulder slabs are below $1e+18$ cm⁻³, where phonon-impurity scattering is negligible against phonon-phonon scattering, so their conductivity is indistinguishable from intrinsic; the model uses 85.3 W/(m·K) at T_{op} . For reference the same fit gives 126 W/(m·K) at 25 °C, which is why a room-temperature parameterisation of this model would have used a figure near 130.

C.4.3 Contact-Region Doping

The contacts are doped far more heavily than the core. The exact level awaits confirmation of the foundry process; $5e+19$ cm⁻³ is the working assumption, and at that density the conductivity reduction is not negligible — the fit gives 65.5 W/(m·K) at T_{op} against 85.3 for the core, a reduction of 23%. This matters more than the modest Si volume fraction suggests: the contacts occupy about fifteen times as many grid

cells as the core and slabs together, and they lie in the lateral path that carries heat outward to the W-fill region.

Only the conductivity is adjusted for doping. Density and specific heat are lattice properties, and $5 \times 10^{19} \text{ cm}^{-3}$ is an atomic fraction near 10^{-3} , far below the uncertainty on Cp itself.

One contact is p-doped and the other n-doped, so the physical cross-section is not laterally symmetric. The model uses the average of the two species. That is not merely tidiness: the solve exploits the lateral mirror symmetry to compute only half the domain (Section 4.1), and a p/n asymmetry would invalidate it. At these concentrations the interpolated p and n values differ by only a few percent, so averaging is an error second order in their difference. Boron does scatter phonons more strongly than phosphorus at equal concentration, consistent with M. Asheghi, K. Kurabayashi, R. Kasnavi and K. E. Goodson, *Thermal conduction in doped single-crystal silicon films*, Journal of Applied Physics **91**, 5079 (2002); that work stops at $3 \times 10^{19} \text{ cm}^{-3}$ and 300 K, so it is cited for the species ordering rather than for the values used here.

C.4.4 Heat Capacities, Silica and Tungsten

Both heat capacities come from the Shomate equations of M. W. Chase Jr., *NIST-JANAF Thermochemical Tables, Fourth Edition*, Journal of Physical and Chemical Reference Data, Monograph 9 (1998), evaluated at T_{op} . For Si(cr) the fit is valid over 298–1685 K and gives 711.8 J/(kg·K) at 298 K rising to 800.3 at T_{op} . For SiO_2 the available fit is for alpha-quartz over 298–847 K, giving 742.0 J/(kg·K) at 298 K rising to 916.2 at T_{op} . The cladding is amorphous rather than crystalline; above roughly 200 K the two forms differ by only a few percent, which is well inside the uncertainty on the silica conductivity discussed next.

The silica thermal conductivity is the weakest-sourced number in the model, and the one the result is most exposed to — silica is 91% of the domain and the dominant thermal resistance. The value used, 1.53 W/(m·K), is the standard 1.38 W/(m·K) for thermal oxide at 300 K carried to T_{op} with a slope of $1.2 \times 10^{-3} \text{ W/(m·K) per K}$, taken from the 300–1000 K trend of P. J. Wray and T. J. Connolly, *Thermal conductivity of clear fused silica at high temperatures*, Journal of Applied Physics **30**, 1702 (1959). No primary source with tabulated values across 300–500 K was obtained. Wray and Connolly's own room-temperature figure is 1.09–1.21 W/(m·K), below the 1.38 handbook value, so the literature spread on this quantity is of order 20% — larger than the temperature correction being applied to it. C.4.6 reports what that does to the answer.

Tungsten conductivity is the modified Hust–Lankford fit of P. Tolias, *Analytical expressions for thermophysical properties of solid and liquid tungsten relevant for fusion applications*, Nuclear Materials and Energy **13**, 42 (2017), valid over 300–3700 K with a mean absolute relative error of 0.39%; it gives 154.0 W/(m·K) at T_{op} . Tungsten specific heat is a handbook value with a weak slope, 137.0 J/(kg·K) at T_{op} . Densities are held at their room-temperature values: linear thermal expansion over this range is about 0.1%.

C.4.5 The W-Fill Composite

The via-fill region is 16% W and 84% SiO_2 by volume. Its conductivity is the volume-weighted arithmetic mean, 25.93 W/(m·K) — a parallel-path approximation, which is the right limit because the vias run through the layer along the direction of heat flow rather than across it. Note that tungsten supplies most of this figure despite the small volume fraction, so the composite conductivity is sensitive to k_{W} and barely at all to k_{SiO_2} .

The volumetric heat capacity needs more care. The heat equation carries $\rho \cdot \text{Cp}$, and for a composite the exact effective value is the **volume** average of $\rho \cdot \text{Cp}$, because both energy and volume are additive:

2,115,122 J/(m³·K). Averaging ρ and C_p separately by volume fraction and multiplying them — a natural-looking shortcut — weights a per-unit-mass quantity by volume and overstated the W-fill thermal mass by 84%. Specific heat has to be weighted by **mass** fraction, and tungsten is 16% of this region by volume but 62% of it by mass. This affects only the transient, not the steady state — and the transient is the subject of this report.

C.4.6 Sensitivity and Remaining Caveats

The property functions are fitted over 25–160 °C. Within that window every property is an interpolation of measured data except the silica conductivity slope. The upper operating limit of 260 °C lies outside the window, but not outside the underlying data: Ohishi reaches 570 K (297 °C) and the Shomate and tungsten fits reach far higher, so only the silica conductivity would be genuinely extrapolated there.

Letting every property track temperature together, the steady-state rise is less sensitive to the operating point than the individual properties are, because the falling conductivities and the rising heat capacities push in opposite directions. Relative to its value at T_{op} it is -5.2% at 105 °C and +1.1% at 160 °C, and -16.6% back at 25 °C. In other words, moving from a 300 K property set to the operating point matters a great deal; drift within the operating range itself does not.

The silica conductivity is a different matter, because its uncertainty is not a temperature question but a literature-spread question. Raising k_{SiO_2} by 9% lowers the steady-state rise by 3.3%, and the full 20% spread between Wray and Connolly's room-temperature value and the handbook figure corresponds to 7.1%. That is the single largest material uncertainty in the model, and the one worth resolving first if these numbers are to be used quantitatively.

Two further approximations remain. Phonon boundary scattering is neglected: at the modelled core cross-section of 370 x 270 nm, phonon mean free paths in Si are comparable to the device dimension, and published values for thin Si layers and nanowires of similar cross-section run lower than the bulk figure used here. This acts in the same direction as every other correction in this appendix — a lower effective conductivity raises the thermal impedance. The Ohishi samples are bulk single crystals, so they carry no such reduction. Second, the temperature dependence within a run is neglected: properties are frozen at T_{op} rather than tracking the local temperature. The pattern-dependent excursions are of order tens of millikelvin, over which the properties are constant to far better than their own uncertainty, so this is consistent; it is the choice of T_{op} itself that matters, not its variation during the transient.

C.5 Uniform Heat Source Distribution

The volumetric heat source Q is assumed constant over the entire waveguide cross-section. In practice the optical absorption is proportional to the local electromagnetic intensity, which follows the guided mode profile — concentrated near the centre of the waveguide and decaying toward the edges. A non-uniform Q would shift heat deposition toward the core, raising the peak temperature relative to the spatially averaged temperature reported here. The mean waveguide temperature (the quantity computed) is insensitive to the in-plane distribution of Q because k_{Si} is high and the waveguide equilibrates quickly. The effect on peak temperature could be a few percent.

C.6 First-Order Time Integration

Backward Euler is first-order accurate in time: the local truncation error per step is $O(dt^2)$, giving a global error of $O(dt)$. The decade-blocking keeps dt small relative to the thermal timescales within each decade, which is sufficient for the smooth thermal responses in this geometry. A Crank-Nicolson scheme (second-order, $O(dt^2)$) would give the same accuracy at half the number of time steps, or equivalent accuracy with fewer decades, at negligible additional implementation cost.

C.7 2D Cross-Section Model — No Axial Conduction

The simulation solves the heat equation in the (x, y) cross-section only, with the z-axis (along the waveguide length) treated as uniform. This is appropriate when the heater length is large compared to the cladding thickness and when end effects can be neglected. For a heater shorter than $\sim 5\ \mu\text{m}$, axial conduction to unheated waveguide sections would provide an additional heat-sinking path, reducing both the steady-state impedance and the rise time. The 2D results reported here represent an upper bound on the thermal impedance for a long, uniformly heated segment.

C.8 Photonic Controller (PCON) Feedback Loop Not Modelled

In a deployed Caribou-MA-DR MRM system, a Photonic Controller (PCON) runs a closed-loop thermal-stabilisation algorithm that continuously adjusts the MRM bias (e.g. via a heater trim or wavelength lock) to compensate for resonance drift. This feedback loop is **not included** in the present model.

Specifically, the model assumes:

Assumption	Consequence
PCON has stabilised the DC thermal operating point before the SSPRQ pattern begins.	The DC steady-state temperature T_{dc} is treated as a fixed offset; only the AC component $\delta T(t)$ is analysed.
PCON does not respond dynamically to pattern-dependent thermal wander during the sequence.	The peak-to-peak δT and PAM-level wander reported here represent the open-loop wander magnitude — an upper bound on what the system would experience in closed-loop operation.
No nonlinear coupling: resonance drift does not feed back onto absorbed optical power.	Absorbed power per PAM level is fixed at its initial DC-balanced value throughout the pattern. This is valid while the wander (91.90% of 1 PAM level peak-to-peak) remains small relative to the linewidth of the resonance.

The practical implication is that the wander values quoted in this report (peak-to-peak $\delta T = 352.07\ \text{mK}$, $23.589\ \text{pm}$, 91.90% of 1 level) are the residual wander seen by the PCON — i.e. the disturbance the controller must reject. Whether the PCON bandwidth is sufficient to suppress this wander, and the eye-closure penalty that results from any unsuppressed residual, are topics for a separate closed-loop analysis and are outside the scope of this report.

C.9 Drop-Port Light Leakage — Not Applicable to DR MRM

In a WDM microring modulator, a small fraction of the circulating optical power exits via the drop port, reducing the net absorbed power in the ring. This would slightly reduce both the steady-state temperature rise and the pattern-dependent thermal wander compared to the values reported here. The Caribou-MA-DR MRM is a **DR (direct-receive) device with no drop port**: all absorbed optical power is deposited as heat in the ring waveguide. This correction does not apply.

C.10 Photocurrent Joule Heating — Small Effect, Neglected

In addition to optical absorption, the reverse-biased MRM junction generates heat from the photogenerated current flowing against the junction voltage: $Q_{\text{elec}} = I_{\text{photo}} \times V_{\text{junction}}$. This term partially offsets the change in absorbed optical power between PAM levels in the following way: at PAM3 (maximum reverse bias), V_{junction} is higher than at PAM0 (minimum reverse bias), while I_{photo} is lower (less optical power circulating in the ring). The PAM3 electrical Joule heating is therefore slightly *larger* than at PAM0, which partially cancels the reduction in optical absorption — the net change in total heat deposited between PAM3 and PAM0 is somewhat smaller than the optical $\text{OMA}_{\text{outer}}$ alone would suggest. This effect is stronger at high background MRM temperatures where photocurrent increases. As a rough bound, it reduces the effective change in heat generated by **no more than ~10 %**, which is within the uncertainty of the present model (grid resolution, bulk Si thermal conductivity, 2D vs. 3D geometry). It is therefore neglected here but should be included in any higher-fidelity analysis aimed at quantitative TDECQ or BER predictions.

Appendix D — Thermal Preconditioning, and Why Cycle 1 Is Not the Answer

D.1 SSPRQ Is DC Balanced

This appendix concerns the DR case only. Over one complete 65,535-symbol cycle the running disparity $RD = \text{sum}(\text{symbol} - 1.5)$ returns to -0.5 symbol-units and the mean symbol is 1.499992 against a balanced 1.5. The residual is the odd symbol count: 65,535 is $2^{16}-1$, so a perfectly balanced sequence of them cannot sum to exactly zero, and half a symbol-unit is the smallest possible miss. **SSPRQ is DC balanced to the limit its length allows.**

Quantity	Value
Running disparity at end of one cycle	-0.5 symbol-units (balanced = 0)
Mean symbol value	1.499992 (balanced = 1.5)
Converged mean delta-T	+0.0102 mK (0.003% of the 352.1 mK pk-pk wander)

So there is **no static thermal offset to accumulate**. The controller has nothing to hold off beyond what C.1 and C.8 already describe, and the converged mean is zero to within a thousandth of the wander it sits inside.

An earlier revision of this report said the opposite. It reported a running disparity of -252 symbol-units and a converged mean of +5.13 mK, and devoted this appendix to how that imbalance accumulated cycle over cycle. That was an artefact of a reconstructed SSPRQ pattern which has since been replaced by the sequence IEEE publishes (F.2). The imbalance was a property of the reconstruction, not of SSPRQ.

D.2 What Preconditioning Is Actually For

Preconditioning is still necessary, for a different reason. The convolution starts from $\text{delta-T} = 0$ — a core at ambient, which is not where a running device sits. The first cycle therefore carries a cold-start transient that has nothing to do with the pattern's balance: the core is filling toward its operating point while the pattern plays, so the mean over cycle 1 is displaced by 4.79 mK (1.4% of the pk-pk wander).

That transient decays with the structure's own time constants rather than accumulating, so it disappears quickly: the mean is within 1% of the pk-pk wander of its final value by cycle 2, within 0.1% by cycle 3, and indistinguishable from converged thereafter. Note the sign — cycle 1 sits *below* the converged mean and rises to meet it, which is the signature of a cold start rather than of a warming bias building up.

The 100 preconditioning cycles are therefore generous rather than marginal: convergence needs of order ten, and 100 were run. Nothing in the reported wander depends on that margin, but it is cheap — the tiled convolution is one FFT — and it removes the question.

D.3 Convergence

Cycle	Mean delta-T (mK)	Note
1	-4.78	Cold start — the convolution begins at ambient
2	-0.46	transient already down to 0.47 mK
3	-0.15	within 0.1% of the pk-pk wander of converged

101	+0.0102	Measurement cycle — periodic steady state
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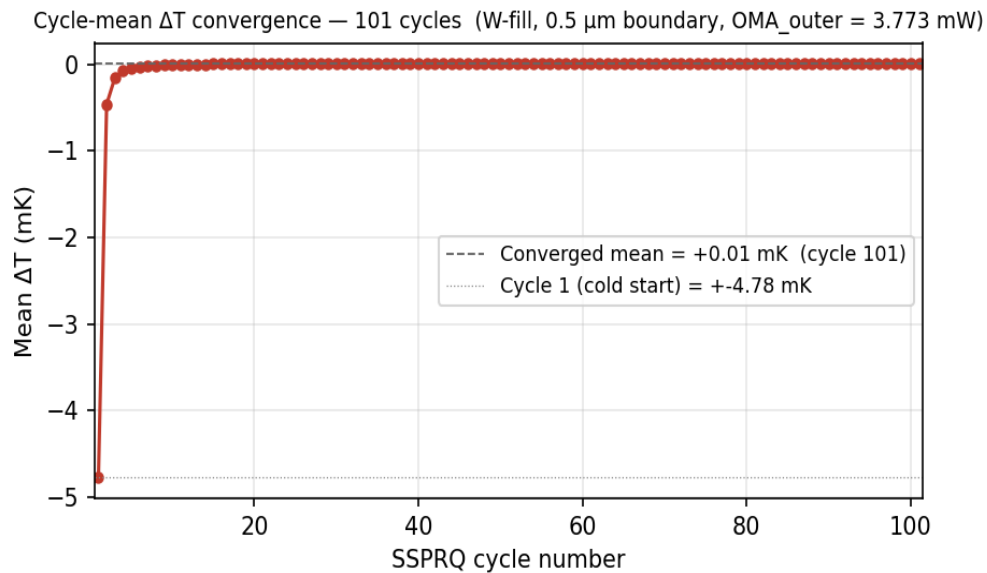


Figure D.1. Mean delta-T per SSQR cycle over all 101 cycles (cycle 1 = cold start; cycle 101 = measurement cycle). The mean rises from -4.78 mK and is flat from about cycle 3 onward at +0.0102 mK. What decays here is the cold-start transient; because the pattern is DC balanced there is no accumulating offset for it to settle onto.

D.4 What Cycle 1 Gets Right — and Wrong

The **AC wander amplitude** — peak-to-peak and sigma — is set by the shape of the pattern and the thermal bandwidth, not by the initial condition, and is essentially identical in cycle 1 and at periodic steady state. What cycle 1 misstates is the **mean**, by 4.79 mK, which is 1.4% of the peak-to-peak wander and 1.25% of one PAM level.

That is small, and it would be tempting to skip the preconditioning on that basis. The reason not to is that the size of the cold-start transient is a property of the structure's time constants, not a constant of the problem: a more thermally isolated ring — the real part is undercut, C.2.2 — settles more slowly and would carry more of it into the measured cycle. Preconditioning costs one FFT and removes the dependence entirely.

The OCI case reaches the same place by a different route. SSQR is exactly DC balanced, by construction rather than by design intent: its blocks 5-8 invert blocks 1-4, so the converged mean is exactly zero. What that construction costs is a strong low-frequency line, which 5.4.2 works through. Both patterns are balanced; only SSQR pays for it with a concentrated tone.

Appendix E — SSPR Pattern Construction and OCI Preconditioning

E.1 SSPR Block Structure

SSPR is specified in 400G CWDM8 MSA 2 km Technical Specifications, Revision 1.1, Section 3.1.2. The generator used here reproduces the normative construction exactly:

Block	Content	Bits
1	PRBS28 (taps 25 and 28), seed 0x0080080	5,437
2	a single one followed by 72 zeros	73
3	PRBS28 (taps 25 and 28), all-ones seed	5,437
4	block 1 less its last 3 bits, differentially coded (zero = change of output, one = no change)	5,434
5 to 8	the bitwise inverse of blocks 1 to 4	16,381
Total	one SSPR cycle	32,762

The specification pins the LFSR convention only implicitly, so it was determined from the normative data rather than assumed: seed 0x0080080 read MSB-first over 28 bits is the sequence the spec gives for the opening of block 1, which fixes the output tap. Of the eight plausible combinations of shift direction, output tap and tap indexing, exactly one reproduces all three of the specification's quoted opening sequences (blocks 1, 3 and 4) — shift left, output the MSB, taps counted from the LSB. All three sequences are asserted inside the generator, so the pattern cannot drift silently if that code is edited.

The specification also permits padding blocks 4 and 8 by 3 bits each to reach 32,768 bits where test equipment requires a power-of-two length. That option is not used here: it exists for instrument limitations a simulation does not have, so the normative 32,762 bits are used.

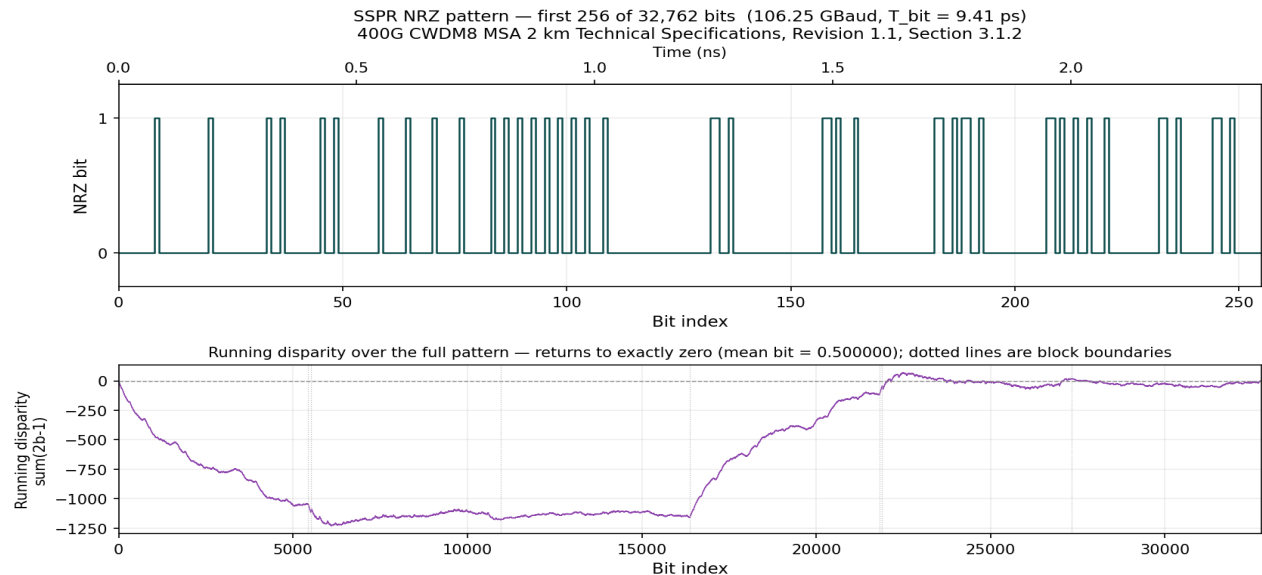


Figure E.1 — SSPR pattern. Upper panel: the first 256 of 32,762 bits at 106.25 GBaud. Lower panel: running disparity over the full pattern, which returns to exactly zero; dotted lines mark the eight block boundaries. The monotonic excursion over the first

half and its mirror over the second are the bitwise inversion of blocks 5 to 8, and are the origin of the 3.24 MHz fundamental discussed in Section 5.4.2.

E.2 OCI Cycle-Mean Convergence

The OCI cycle period (308.35 ns) is comparable to τ_{90} (578.7 ns), as is the DR cycle (617 ns) against the same τ_{90} , so neither case is in the $T_{\text{cycle}} \gg \tau_{90}$ limit and both were preconditioned over 100 cycles. For OCI the converged mean is a stronger statement than for DR: because SSPR is exactly DC balanced, the cycle mean converges to exactly zero rather than to a pattern-determined offset. The cold-start cycle sits at +10.178 mK — the transient of starting from a uniform temperature, not a property of the pattern — and decays to +0.000 mK by cycle 101.

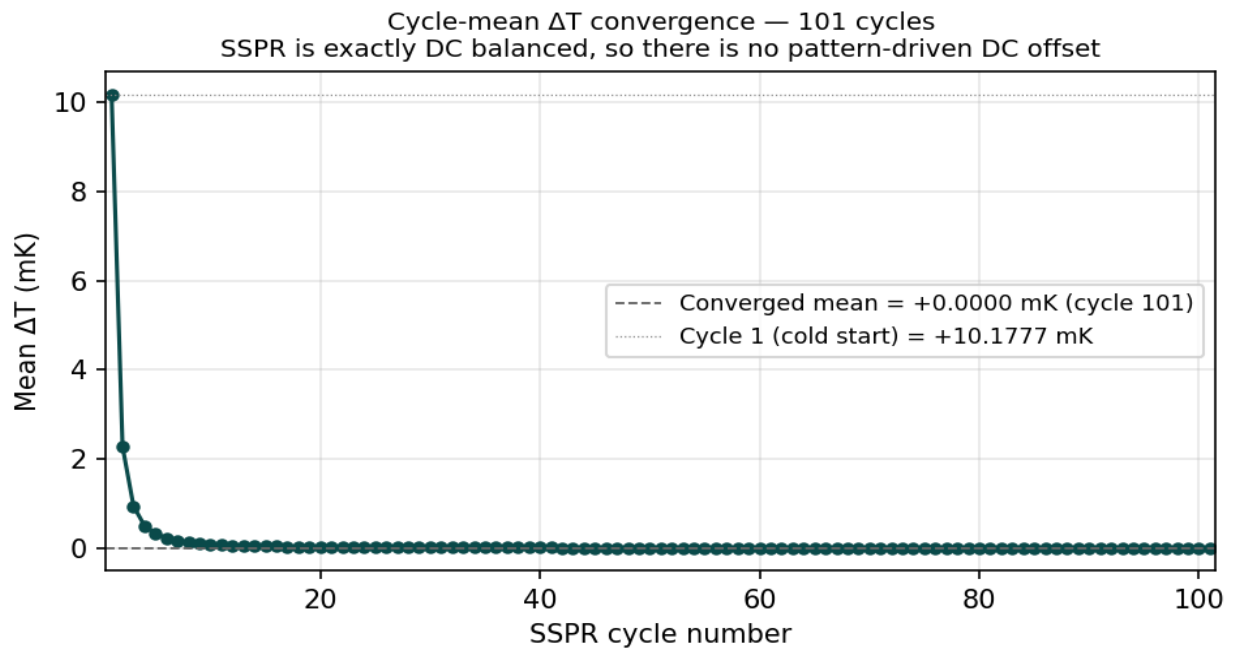


Figure E.2 — OCI cycle-mean delta-T over 100 preconditioning cycles plus the measurement cycle. The approach to zero is the decay of the cold-start transient; there is no DC imbalance for it to converge onto. Figure D.1 shows the same behaviour for DR, which converges to +0.010 mK — both patterns are DC balanced, so in both cases what decays is the cold start and not an accumulating offset.

Appendix F — References

Chicago style, bibliography format. **Copies of the sources that supply numbers to this model are in the project Google Drive folder alongside the published report.** The standard texts cited in F.3 for background are not held there, being widely available. Where a source is used for a specific number rather than for general background, the note after the entry says which.

F.1 Material Properties

Asheghi, M., K. Kurabayashi, R. Kasnavi, and K. E. Goodson. “Thermal Conduction in Doped Single-Crystal Silicon Films.” *Journal of Applied Physics* 91, no. 8 (2002): 5079–88.

Cited in C.4.3 for the ordering of the two dopant species — boron scatters phonons more strongly than phosphorus at equal concentration. Its measurements stop at $3 \times 10^{19} \text{ cm}^{-3}$ and 300 K, so no value from it is used.

Chase, Malcolm W., Jr. *NIST-JANAF Thermochemical Tables*. 4th ed. Journal of Physical and Chemical Reference Data, Monograph 9. Washington, DC: American Chemical Society; Woodbury, NY: American Institute of Physics, 1998.

Source of both heat capacities (C.4.4), through the Shomate equations for Si(cr) over 298–1685 K and for SiO₂ alpha-quartz over 298–847 K, evaluated at T_{op} : $C_{p_{\text{Si}}} = 800.3$ and $C_{p_{\text{SiO}_2}} = 916.2 \text{ J/(kg}\cdot\text{K)}$.

Ohishi, Yuji, Jun Xie, Yoshinobu Miyazaki, Yusufu Aikebaier, Hiroaki Muta, Ken Kurosaki, Shinsuke Yamanaka, Noriyuki Uchida, and Tetsuya Tada. “Thermoelectric Properties of Heavily Boron- and Phosphorus-Doped Silicon.” *Japanese Journal of Applied Physics* 54, no. 7 (2015): 071301.

The primary source for silicon thermal conductivity (C.4.2). Table IV gives kappa at 300 K and 570 K for six single-crystal samples spanning 10^{18} – 10^{20} cm^{-3} in both dopants; interpolation in log(doping) and then in temperature gives 85.3 W/(m·K) for the core and slab and 65.5 for the contacts at T_{op} .

Tolias, Panagiotis. “Analytical Expressions for Thermophysical Properties of Solid and Liquid Tungsten Relevant for Fusion Applications.” *Nuclear Materials and Energy* 13 (2017): 42–57.

Source of the tungsten thermal conductivity (C.4.4) through the modified Hust–Lankford fit, valid 300–3700 K: $k_{\text{W}} = 154.0 \text{ W/(m}\cdot\text{K)}$ at T_{op} .

Wray, P. J., and T. J. Connolly. “Thermal Conductivity of Clear Fused Silica at High Temperatures.” *Journal of Applied Physics* 30, no. 11 (1959): 1702–5.

Source of the temperature slope for the silica conductivity (C.4.4). The 300 K anchor of 1.38 W/(m·K) is instead a widely used handbook figure for thermal oxide with no single primary source; Wray and Connolly's own room-temperature value is 1.09–1.21. That disagreement is why this property carries the largest uncertainty in the model.

F.2 Test Patterns

400G CWDM8 MSA Group. *400G CWDM8 2 km Optical Interface Technical Specifications*. Revision 1.1. February 13, 2018. cwdm8-msa.org.

Its section 3.1.2 defines SSPR, the 32,762-bit NRZ pattern used for the Caribou-MA-OCI case (Section 5.4 and Appendix E.1).

Institute of Electrical and Electronics Engineers. *IEEE Standard for Ethernet*. IEEE Std 802.3-2022. New York: IEEE, 2022.

Clause 120.5.11.2.3 defines SSPRQ, the 65,535-symbol PAM4 pattern used for the Caribou-MA-DR case (Section 4.5), and 120.5.7.1 the Gray mapping. **The pattern is not reconstructed from the clause text.** It is read verbatim from the symbol sequence IEEE publishes alongside the standard, which that clause's NOTE points to, held in this project as reference/Clause_120_SSPRQ_sequence.csv and checked by sha256 on every load. An earlier version of this work did reconstruct it, and differed from the published sequence in 74.7% of symbols.

F.3 Thermal Frequency-Domain Methods

Background for the frequency-domain treatment in 4.3. **No number in this model comes from any of these.** They are cited to establish that describing heat conduction by a transfer function against frequency is standard practice with settled terminology, not a construction of this report — a reasonable question to ask of a quantity that is more often met as a time-domain transient.

Carslaw, H. S., and J. C. Jaeger. *Conduction of Heat in Solids*. 2nd ed. Oxford: Clarendon Press, 1959.

The classical treatment, including the response of a solid to periodic heating — the frequency-domain view used in 4.3. Cited for the concept only.

Cahill, David G. “Thermal Conductivity Measurement from 30 to 750 K: The 3-omega Method.” *Review of Scientific Instruments* 61, no. 2 (1990): 802–8.

The 3-omega technique is a frequency-domain thermal measurement: the sample is driven at one frequency and its thermal response read out across a sweep. It is the standard method for thin-film thermal conductivity, and is the clearest evidence that thermal frequency response is an established experimental quantity and not only a modelling device.

JEDEC Solid State Technology Association. *JESD51-14: Transient Dual Interface Test Method for the Measurement of the Thermal Resistance Junction-to-Case of Semiconductor Devices with Heat Flow through a Single Path*. Arlington, VA: JEDEC, November 2010.

The industry standard for transient thermal impedance Z_{th} , the quantity 4.2 and 4.3 show in the time and frequency domains respectively. Cited for terminology: *thermal impedance* unqualified normally means the time-domain transient, which is why this report says *thermal frequency response* for the transfer function and reserves *DC thermal impedance* for H_{dc} .

F.4 Internal Sources

Lightmatter. “Caribou_OMA_rollup_2026-08-31.” Internal spreadsheet.

Source of both optical modulation amplitudes: use case 80, cell H36 for the DR OMA_{outer} of 5.767 dBm, and use case 70, cell H40 for the OCI OMA of 3.793 dBm.

Not from the literature. Three quantities in this model have no external source and rest on device or process information: the ring radii, the cross-section geometry, and the contact doping concentration. The

last of these is a working assumption of $5 \times 10^{19} \text{ cm}^{-3}$ pending confirmation of the foundry process, and C.4.3 gives its effect.

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